EXAM PACK

EEE337

Power System II



Exam Pack

Session 2020–21 *Solutions and formula sheet of the 3rd Year 2nd Semester Final Examination, 2024*

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Question Paper Solutions 2020–21 — pages 3--60 — 58 pages

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Prepared by Nur Mohammad

October 2026

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PAST PAPER SOLUTIONS

EEE337

Power System II

Question Paper Solutions

Session 2020–21

3rd Year 2nd Semester Final Examination, 2024

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

EACH ANSWER

Theory → Answer → Keyword Explanation → Notes and Remarks → Final Concise Answer

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October 2026

BACK OF THE COVER

The paper at a glance

3 h	100
Total time	Total marks

Time = marks × 1.8 min (180 min for 100 marks). Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
PART A				
Q1	(a) Definition of Inductance	1	2 min	
	(b) Inductive Reactance of a Bundled Pheasant Line	4	7 min	
	(c) Capacitance of a Three-Phase Line with Unsymmetrical Spacing	10	18 min	☐
	(d) Mutual Inductance between a Power Line and a Telephone Line in One Plane	10	18 min	
Q2	(a) Corona Effect	1	2 min	
	(b) Transposed Transmission Line	4	7 min	☐
	(c) Inductance of Composite Conductor Lines	10	18 min	
	(d) Maximum and Minimum Stress in a 33 kV Single-Core Cable	10	18 min	
Q3	(a) Skin Effect	1	2 min	
	(b) FACTS Devices and Their Applications	4	7 min	☐
	(c) Dielectric Stress in a Single-Core Cable	10	18 min	
	(d) Charging Current of a Belted Three-Core Cable	10	18 min	
PART B				
Q4	(a) Critical Disruptive Voltage and Its Significance	5	9 min	
	(b) String Efficiency and the Methods of Improving It	10	18 min	☐
	(c) Insulation Thickness and Operating Voltage from the Stresses	10	18 min	
Q5	(a) Causes of Insulator Failure	5	9 min	
	(b) Locating an Earth Fault by the Varley Loop Test	10	18 min	☐
	(c) Height of the Supports from the Sag	10	18 min	
Q6	(a) Power Angle and the Equal Area Criterion	5	9 min	
	(b) Sudden Increase in Mechanical Input on the Power-Angle Diagram	10	18 min	☐
	(c) Transient EMF and Maximum Power, Healthy and with a Mid-Line Fault	10	18 min	

Formulas and constants follow V.K. Mehta, Principles of Power System (line constants, corona, insulators, sag, cables), Grainger and Stevenson, Power System Analysis (ACSR conductor data, bundled lines, stability) and Hingorani and Gyugyi, Understanding FACTS (FACTS controllers).

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
3rd Year 2nd Semester Final Examination, 2024
Course Code: EEE 337, Session: 2020-2021
Course Title: Power System II

Full Marks: 100 **Credit: 3.0** **Exam Duration: 3 Hours**

Answer any two questions from each part

Part A

1. (a) Define Inductance with the required expression. 1
   (b) Each conductor of the bundled-conductor line shown in Fig.1 is ACSR, 1,272,000-cmil Pheasant. Find the inductive reactance in ohms per km (and per mile) per phase for d=45cm. 4
   (c) Derive the expression of Capacitance of a three phase line with unsymmetrical spacing. 10
   (d) A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line. 10

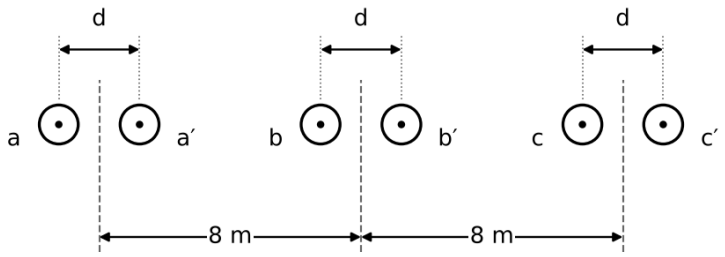


Fig. 1 of the paper (redrawn): figure for Q-1(b)

2. (a) Define Corona Effect. 1
   (b) Define Transposed Transmission line. Explain the reason behind Transpose. 4
   (c) Derive the expression of Inductance of composite conductor lines. 10
   (d) A 33 kV single core cable has a conductor diameter of 1 cm and a sheath of inside diameter 4 cm. Find the maximum and minimum stress in the insulation. 10

3. (a) Define Skin Effect. 1
   (b) Describe different FACTS devices and applications. 4
   (c) Derive the expression of dielectric stress in a single core cable. 10
   (d) The capacitances of a 3-phase belted cable are $12 \cdot 6$ μF between the three cores bunched together and the lead sheath and $7 \cdot 4$ μF between one core and the other two connected to sheath. Find the charging current drawn by the cable when connected to 66 kV, 50 Hz supply. 10

Part B

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4. (a) Define critical disruptive voltage. Write down its significance. 5

   (b) What is String Efficiency? Discuss the methods of improving String Efficiency. 10

   (c) The maximum and minimum stresses in the dielectric of a single core cable are 40 kV/cm (r.m.s.) and 10 kV/cm (r.m.s.) respectively. If the conductor diameter is 2 cm, find : (i) thickness of insulation (ii) operating voltage 10

5. (a) What are the causes of insulation failure? 5

   (b) Varley loop test is performed to locate an earth fault on a 20 km long cable. The resistance per km of the single conductor is 20 ohm. The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 ohm. The fixed resistors have equal values. Calculate the distance of the fault from the test end. 10

   (c) A 132 KV transmission line has the following data: Weight of the conductor= 680 kg/km, Length of span= 260 m, Ultimate strength= 3100 kg, Safety Factor= 2. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 meters. 10

6. (a) Define Power Angle. Briefly explain Equal Area Criterion. 5

   (b) Explain what happens for sudden increase in mechanical input to generator using Pe-δ diagram. 10

   (c) The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus (|V|=1.0 pu), with the generator terminal voltage (|Vt|=1.0 pu). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (a) System healthy (b) One line shorted (3 phase) in the middle. 10

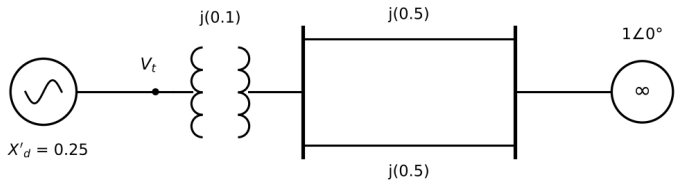


Fig. 2 of the paper (redrawn; the same figure as in the 2021-22 paper): figure for Q-6(c)

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1. (a) Define Inductance with the required expression. Marks: 1

   (b) Each conductor of the bundled-conductor line shown in Fig.1 is ACSR, 1,272,000-cmil Pheasant. Find the inductive reactance in ohms per km (and per mile) per phase for d=45cm. Marks: 4

   (c) Derive the expression of Capacitance of a three phase line with unsymmetrical spacing. Marks: 10

   (d) A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line. Marks: 10

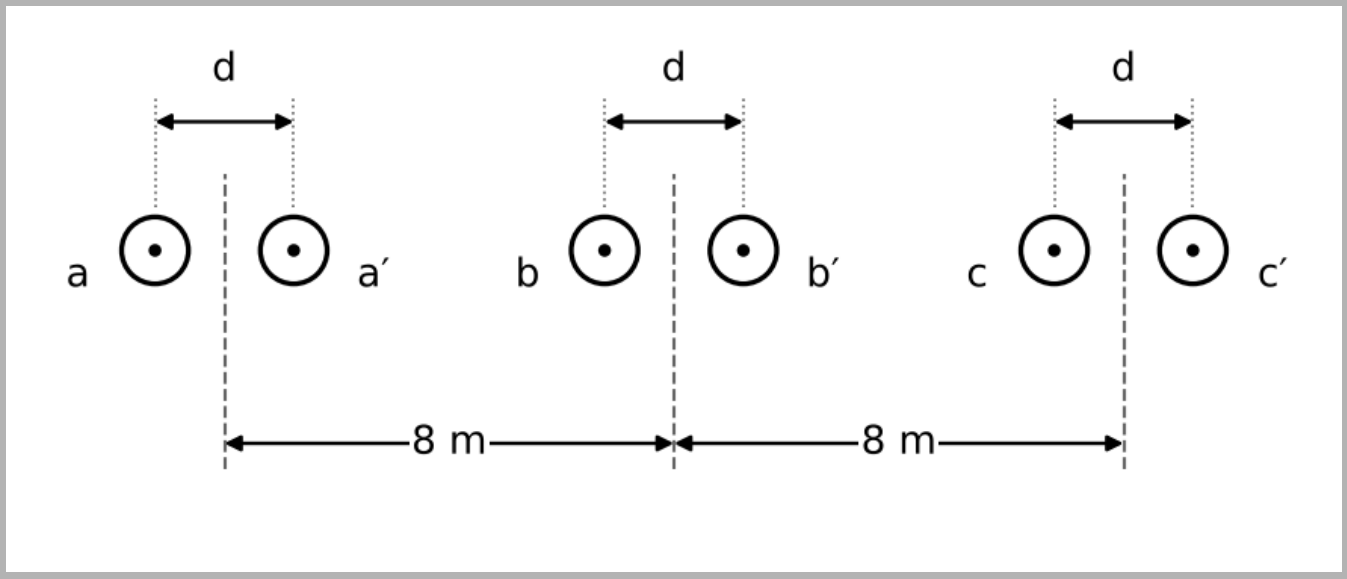


Figure 1: Fig. 1 of the paper (redrawn): figure for Q-1(b)

1.(a) Define Inductance with the required expression. Marks: 1

1. Theory

1. A current makes a magnetic field (Core Idea)

When a current flows in a wire, it makes a **magnetic field** around the wire. The field lines are circles around the wire, like rings on a finger. The total amount of field passing through a surface is the **magnetic flux** ϕ, measured in weber (Wb).

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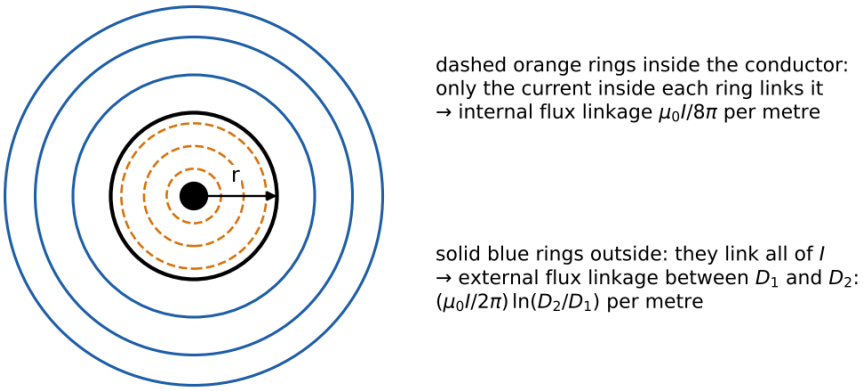


Figure 2: Flux of one conductor carrying current I out of the page: dashed orange rings inside the conductor, solid blue rings outside (reused from the 2021-22 paper)

Reading the figure. The current flows out of the page (the dot). Every ring of flux surrounds the current. Rings inside the conductor (dashed orange) surround only part of the current; rings outside (solid blue) surround all of it.

2. Flux linkage

A ring of flux that goes around the current once is said to **link** it once. If a coil has N turns and a flux ϕ passes through all of them, the flux is linked N times. The **flux linkage** is

$$\psi = N\phi \quad \text{weber-turns (Wb-T)}$$

For one straight wire, $N = 1$, but the rings inside the wire surround only a fraction of the current, so they count as a fraction of a linkage.

3. Why the flux linkage matters

In an AC circuit the current changes all the time, so the flux linkage changes too. By Faraday's law, a changing flux linkage makes a voltage $e = d\psi/dt$ that **opposes** the change of current. This is the voltage drop we feel as the line's reactance. A larger flux linkage per ampere means a larger opposing voltage.

2. Answer

Inductance is the flux linkage of a circuit per ampere of the current that produces it. It measures how strongly the circuit opposes a change of its current:

$$L = \frac{\psi}{I} = \frac{N\phi}{I} \quad \text{henry (H)}$$

where ψ is the flux linkage (Wb-T), N the number of turns, ϕ the flux (Wb) and I the current (A). The voltage induced is $e = L\,\dfrac{di}{dt}$. For a transmission line, L is given per metre (H/m) or per km, and its reactance is $X_L = 2\pi f L$.

3. Keyword Explanation

1. Flux linkage

The flux multiplied by the number of turns (or the fraction of the current) it surrounds, in weber-turns.

2. Henry

One henry is the inductance in which a current change of 1 A/s induces 1 V; equivalently 1 Wb-T per ampere.

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1.(b) Each conductor of the bundled-conductor line shown in Fig.1 is ACSR, 1,272,000-cmil Pheasant. Find the inductive reactance in ohms per km (and per mile) per phase for d=45cm. Marks: 4

1. Theory

1. Flux linkage of one long round conductor (Core Idea)

From 1(a), a conductor of radius r carrying current I is surrounded by rings of flux.

- **Inside** the conductor, a ring of radius x surrounds only the current inside it, Ix^2/r^2 (the current is spread evenly over the cross-section). Adding all these part linkages gives

$$\psi_{int} = \frac{\mu_0 I}{8\pi} \text{ Wb-T/m}$$

- **Outside**, the field at distance x is $H = \dfrac{I}{2\pi x}$ and each ring links the whole current. The rings between distances D_1 and D_2 from the centre give

$$\psi_{12} = \frac{\mu_0 I}{2\pi} \ln \frac{D_2}{D_1} \text{ Wb-T/m}$$

Here $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability of air, so $\dfrac{\mu_0}{2\pi} = 2 \times 10^{-7}$.

2. The geometric mean radius (GMR)

Adding the inside part to the outside part up to a distance D:

$$\psi = \frac{\mu_0 I}{2\pi}\left(\frac{1}{4} + \ln\frac{D}{r}\right) = 2\times10^{-7} I \ln\frac{D}{re^{-1/4}} = 2\times10^{-7} I \ln\frac{D}{D_s}$$

$D_s = re^{-1/4} = 0.7788\,r$ for a solid round wire. D_s is the **geometric mean radius** (GMR): the radius of an imaginary thin tube that has the same inductance as the real conductor. With it, the inside of the conductor needs no separate treatment.

A real line conductor is **stranded**, and an ACSR conductor has a steel core that carries almost no current. Its GMR is therefore not exactly $0.7788\,r$; it is measured and listed in conductor tables.

3. ACSR and the name "1,272,000-cmil Pheasant"

- **ACSR** (aluminium conductor, steel reinforced) has layers of aluminium strands around a core of steel strands. The aluminium carries the current; the steel gives strength.
- **cmil** (circular mil) is an area unit: the area of a circle 1 mil (0.001 inch) in diameter. 1,272,000 cmil is the aluminium area, about 6.445×10^{-4} m^2 (645 mm^2).
- **Pheasant** is the code name of this size. Grainger and Stevenson's Table A.3 lists it with an outside diameter of 1.382 in and a **GMR of** $D_s = 0.0466$ **ft** (0.0142 m).

4. Three-phase line with unequal spacing, transposed

For three conductors a, b, c with balanced currents ($I_a + I_b + I_c = 0$), the flux linkage of a is

$$\psi_a = 2\times10^{-7}\left[I_a \ln\frac{1}{D_s} + I_b \ln\frac{1}{D_{ab}} + I_c \ln\frac{1}{D_{ac}}\right]$$

With unequal distances, each phase would get a different inductance. The line is therefore **transposed** (each conductor takes each position for one third of the length; see 2(b)). Averaging over the three sections gives the same inductance for all phases:

$$L = 2\times10^{-7}\ln\frac{D_{eq}}{D_s} \text{ H/m per phase}, \qquad D_{eq} = \sqrt[3]{D_{ab}D_{bc}D_{ca}}$$

D_{eq} is the **equivalent spacing** (geometric mean distance between the phases).

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5. Bundled conductors

At high voltages each phase is often made of **two or more sub-conductors** held a short distance d apart by spacers. This is a **bundled conductor**. It lowers the electric field at the conductor surface (less corona, see 2(a)) and lowers the inductance.

A two-conductor bundle is a composite conductor (2(c)) with two equal strands. Its GMR is the 4th root of the four distances inside it: each sub-conductor to itself (D_s) and to the other (d):

$$D_s^b = \sqrt[4]{D_s \cdot d \cdot d \cdot D_s} = \sqrt{D_s\, d}$$

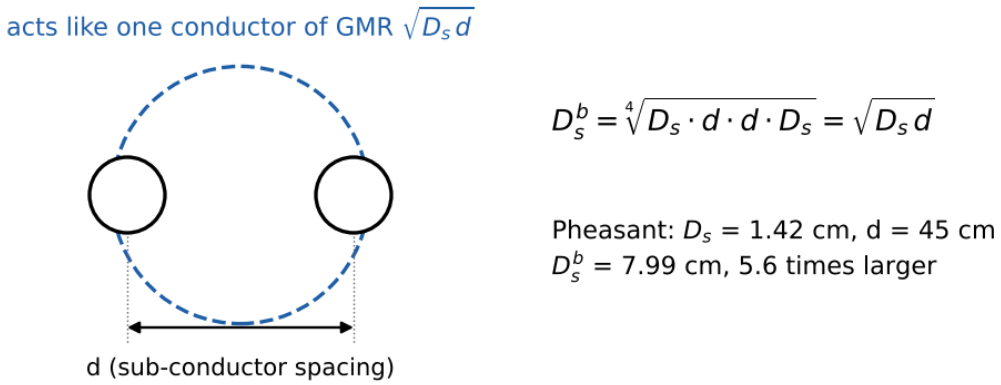


Figure 3: A two-conductor bundle acts like one conductor whose GMR is the square root of Ds times d

Reading the figure. The two sub-conductors (circles) are d apart. Seen from far away, they behave like one fat conductor (dashed blue circle) whose GMR is $\sqrt{D_s d}$: for Pheasant with $d = 45$ cm this is 7.99 cm, 5.6 times the GMR of one conductor. A larger GMR means a smaller $\ln(D_{eq}/D_s)$ and so a smaller inductance.

Because the bundles are far apart compared with d, the distances between phases are measured between the **centres** of the bundles. The reactance per phase is then

$$X_L = 2\pi f L = 2\pi f \times 2 \times 10^{-7} \ln \frac{D_{eq}}{D_s^b} \ \ \Omega/\text{m}$$

2. Answer

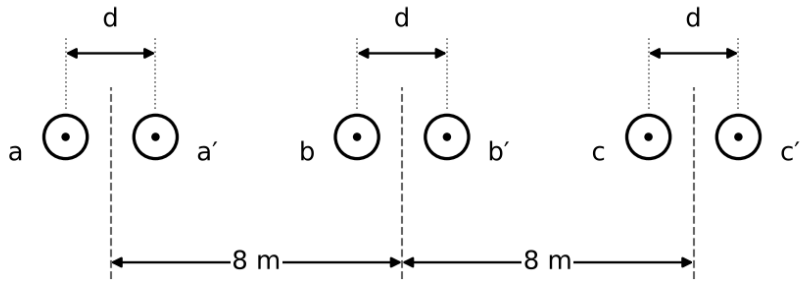


Figure 4: Repeated from the question: Fig. 1 of the paper (redrawn): figure for Q-1(b)

Given:

- Pheasant ACSR, GMR of one conductor $D_s = 0.0466$ ft $= 0.0466 \times 0.3048 = 0.014204$ m (Grainger and Stevenson, Table A.3)

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- Sub-conductor spacing $d = 45$ cm $= 0.45$ m
- Spacing of the bundle centres: $D_{ab} = D_{bc} = 8$ m, $D_{ca} = 16$ m (Fig. 1)

Assumption: the paper gives no frequency. This is Example 4.6 of Grainger and Stevenson, which uses **60 Hz**, so $f = 60$ Hz (the 50 Hz values are in the Notes). The line is taken as transposed.

Step 1: GMR of the bundle

$$D_s^b = \sqrt{D_s\, d} = \sqrt{0.014204 \times 0.45} = 0.07995 \text{ m}$$

Step 2: Equivalent spacing

$$D_{eq} = \sqrt[3]{8 \times 8 \times 16} = \sqrt[3]{1024} = 10.079 \text{ m}$$

Step 3: Inductance per phase

$$\frac{D_{eq}}{D_s^b} = \frac{10.079}{0.07995} = 126.07, \qquad \ln 126.07 = 4.8369$$

$$L = 2 \times 10^{-7} \times 4.8369 = 9.674 \times 10^{-7} \text{ H/m} = 0.9674 \text{ mH/km}$$

Step 4: Inductive reactance per phase

$$X_L = 2\pi \times 60 \times 0.9674 \times 10^{-3} = 0.3647 \ \Omega/\text{km}$$

$$X_L = 0.3647 \times 1.609 = 0.5869 \ \Omega/\text{mi}$$

$$\boxed{X_L = 0.365 \ \Omega \text{ per km} = 0.587 \ \Omega \text{ per mile, per phase}}$$

Final Answer: $D_s^b = 0.0800$ m, $D_{eq} = 10.08$ m; $X_L = 0.365\ \Omega/\text{km} = 0.587\ \Omega/\text{mi}$ per phase (60 Hz).

3. Notes and Remarks

- **At 50 Hz** the reactance is 5/6 of these values: $X_L = 0.304\ \Omega/\text{km} = 0.489\ \Omega/\text{mi}$ per phase.
- **The handwritten 0.0244 m on the scan:** if it is taken as D_s, then $D_s^b = 0.1048$ m and $X_L = 0.344\ \Omega/\text{km}$ (0.554 Ω/mi) at 60 Hz, or 0.287 Ω/km at 50 Hz. The book's table value 0.0466 ft is used above.
- **Gain from bundling:** one Pheasant per phase at the same D_{eq} would give $X_L = 2\pi \times 60 \times 2 \times 10^{-7} \ln(10.079/0.014204) = 0.495\ \Omega/\text{km}$. The bundle lowers the reactance by about 26 %.
- **Exact distances:** the four sub-conductor distances between two neighbouring bundles are 8, 8.45, 7.55 and 8 m; their geometric mean is 7.994 m instead of 8 m. This changes D_{eq} to 10.073 m and leaves $X_L = 0.365\ \Omega/\text{km}$.
- **Why not $0.7788\,r$?** With $r = 1.382/2$ in, $0.7788\,r = 0.0448$ ft, close to but not equal to the table's 0.0466 ft, because the conductor is stranded and the steel core carries little current. Always use the table GMR for ACSR.
- A similar bundled line (two 30 mm conductors, 0.5 m apart) is solved in 7(a) of the 2021-22 paper, as extra reading.

1.(c) Derive the expression of Capacitance of a three phase line with unsymmetrical spacing. Marks: 10

1. Theory

1. A line is a capacitor (Core Idea)

Two conductors separated by an insulator form a **capacitor**. The wires of an overhead line are conductors, and the air between them is the insulator. When the line is at a voltage, a **charge** sits on each wire, $+q$ coulomb per metre on one and $-q$ on another. **Capacitance** is the charge per volt:

$$C = \frac{q}{V} \ \text{F/m}$$

Because the voltage is AC, the charges change all the time, and the current that carries them in and out is the line's **charging current**. It flows even when no load is connected.

2. The field and the voltage near a charged wire

A long straight wire with charge q per metre sends its electric field straight outwards. By Gauss's law, the field spreads over a cylinder of surface $2\pi x$ per metre, so at distance x

$$E = \frac{q}{2\pi\varepsilon_0\, x}$$

where $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m is the permittivity of air. The voltage difference between two points P_1 and P_2 at distances D_1 and D_2 from the wire is the field added up from P_1 to P_2:

$$V_{12} = \int_{D_1}^{D_2} \frac{q}{2\pi\varepsilon_0 x}\, dx = \frac{q}{2\pi\varepsilon_0} \ln \frac{D_2}{D_1}$$

The point nearer the positive charge is at the higher voltage.

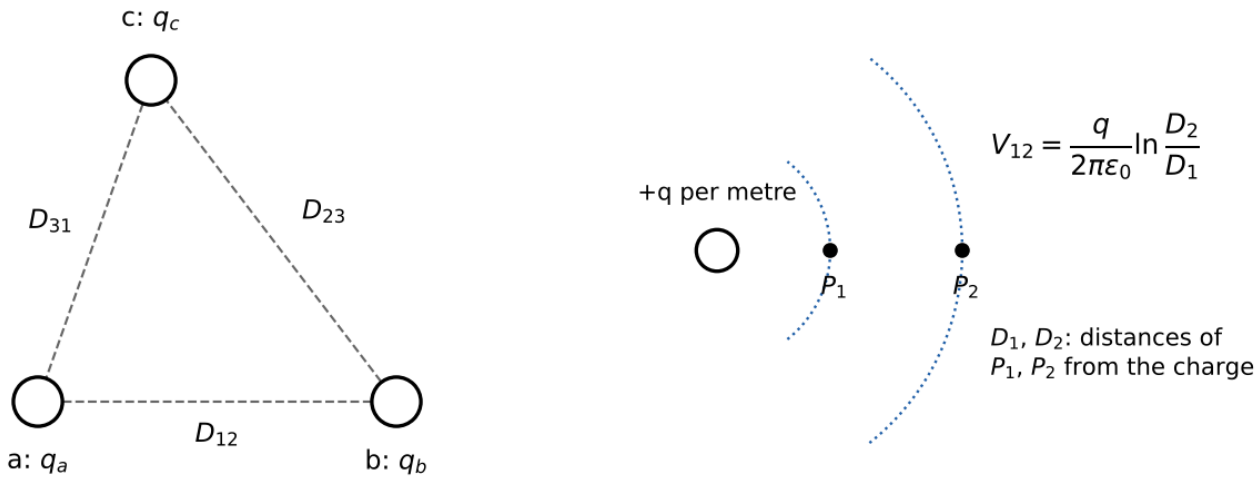


Figure 5: Left: three conductors with charges qa, qb, qc at unequal distances. Right: the voltage between two points near a line charge (reused from the 2021-22 paper)

Reading the figure. On the left, the three line conductors a, b, c sit at the corners of an unequal triangle with sides D_{12}, D_{23}, D_{31}. On the right, the point P_1 (nearer the charge) is at a higher voltage than P_2 by $\frac{q}{2\pi\varepsilon_0} \ln \frac{D_2}{D_1}$.

3. Several charged wires: superposition

When several charged wires are present, the voltage between two of them is the **sum** of the contributions of every charge. For example, between conductors a and b, the charge q_a contributes $\frac{q_a}{2\pi\varepsilon_0} \ln \frac{D_{ab}}{r}$ (from the surface of a, at distance r, out to b, at distance D_{ab}).

4. Why unsymmetrical spacing needs transposition

If the three conductors are at the corners of an equilateral triangle, each phase sees the same distances, and the capacitance to neutral is $C_n = 2\pi\varepsilon_0 / \ln(D/r)$. If the spacing is unequal (for example, side by side on a cross-arm), each phase would have a different capacitance and a different charging current. The line is therefore **transposed**: each conductor takes each of the three positions for one third of the length (2(b)).

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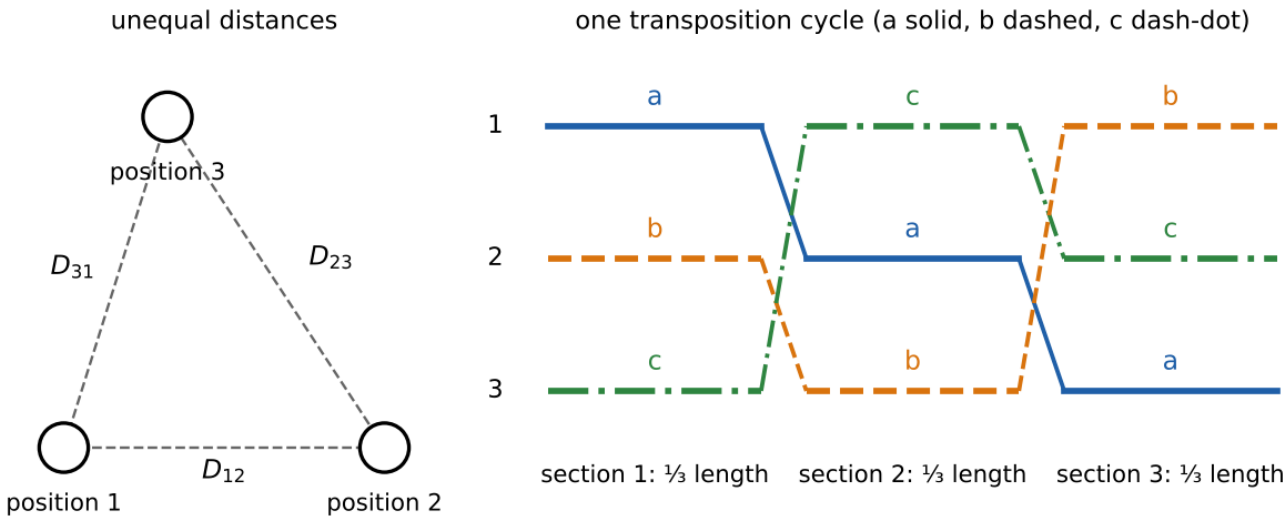


Figure 6: Left: three positions at unequal distances D12, D23, D31. Right: one transposition cycle; conductor a (solid blue), b (dashed orange) and c (dash-dot green) each take every position for one third of the length (reused from the 2021-22 paper)

Reading the figure. Follow conductor a (solid blue): it is at position 1 in section 1, position 2 in section 2 and position 3 in section 3. Conductors b and c also visit every position once, so on average all three conductors have the same surroundings.

2. Answer

Set-up. Three conductors a, b, c, each of radius r, occupy positions 1, 2, 3 with distances D_{12}, D_{23}, D_{31} (figure above). The line is transposed. The charges per metre are q_a, q_b, q_c, with

$$q_a + q_b + q_c = 0$$

for balanced conditions. We assume each conductor keeps the same charge in all three sections (the usual textbook assumption).

Voltage of a point due to a line charge. Between distances D_1 and D_2 from a wire of charge q:

$$V = \frac{q}{2\pi\varepsilon_0} \ln \frac{D_2}{D_1}$$

Voltage V_{ab} in each transposition section (adding the contributions of q_a, q_b and q_c):
Section 1 (a at 1, b at 2, c at 3):

$$V_{ab1} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln \frac{D_{12}}{r} + q_b \ln \frac{r}{D_{12}} + q_c \ln \frac{D_{23}}{D_{31}}\right]$$

Section 2 (a at 2, b at 3, c at 1):

$$V_{ab2} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln \frac{D_{23}}{r} + q_b \ln \frac{r}{D_{23}} + q_c \ln \frac{D_{31}}{D_{12}}\right]$$

Section 3 (a at 3, b at 1, c at 2):

$$V_{ab3} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln \frac{D_{31}}{r} + q_b \ln \frac{r}{D_{31}} + q_c \ln \frac{D_{12}}{D_{23}}\right]$$

Average over the transposition cycle (the three sections are of equal length):

$$V_{ab} = \frac{V_{ab1} + V_{ab2} + V_{ab3}}{3} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln \frac{\sqrt[3]{D_{12}D_{23}D_{31}}}{r} + q_b \ln \frac{r}{\sqrt[3]{D_{12}D_{23}D_{31}}}\right]$$

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The q_c terms cancel, because $\frac{D_{23}}{D_{31}} \cdot \frac{D_{31}}{D_{12}} \cdot \frac{D_{12}}{D_{23}} = 1$ and $\ln 1 = 0$. Write $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$:

$$V_{ab} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln\frac{D_{eq}}{r} + q_b \ln\frac{r}{D_{eq}}\right] \qquad (1)$$

In the same way, for V_{ac}:

$$V_{ac} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln\frac{D_{eq}}{r} + q_c \ln\frac{r}{D_{eq}}\right] \qquad (2)$$

Add (1) and (2) and use $q_b + q_c = -q_a$:

$$V_{ab} + V_{ac} = \frac{1}{2\pi\varepsilon_0}\left[2q_a \ln\frac{D_{eq}}{r} - (q_b + q_c)\ln\frac{D_{eq}}{r}\right] = \frac{3q_a}{2\pi\varepsilon_0}\ln\frac{D_{eq}}{r} \qquad (3)$$

Relate to the phase voltage. For balanced three-phase voltages, $V_{ab} = \sqrt{3}\,V_{an}\angle 30°$ and $V_{ac} = -V_{ca} = \sqrt{3}\,V_{an}\angle -30°$. Their sum is

$$V_{ab} + V_{ac} = \sqrt{3}\,V_{an}\,(\angle 30° + \angle -30°) = \sqrt{3}\,V_{an} \times 2\cos 30° = 3V_{an}$$

Putting this in (3):

$$3V_{an} = \frac{3q_a}{2\pi\varepsilon_0}\ln\frac{D_{eq}}{r} \quad \Rightarrow \quad V_{an} = \frac{q_a}{2\pi\varepsilon_0}\ln\frac{D_{eq}}{r}$$

Capacitance to neutral (charge per volt to neutral):

$$\boxed{C_n = \frac{q_a}{V_{an}} = \frac{2\pi\varepsilon_0}{\ln\left(\sqrt[3]{D_{12}D_{23}D_{31}}/r\right)} \ \text{F/m per phase}}$$

With $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m, this is $C_n = \dfrac{0.0556}{\ln(D_{eq}/r)}$ μF/km. By the same steps C_b and C_c come out equal to C_a: transposition makes the three phases alike. For symmetrical spacing $D_{eq} = D$ and the result becomes $C_n = 2\pi\varepsilon_0/\ln(D/r)$.

Charging current per phase: $I_c = 2\pi f\, C_n\, V_{ph}$.

3. Keyword Explanation

1. Capacitance to neutral

The capacitance between one conductor and the neutral point of a three-phase system; the charging current per phase is $\omega C_n V_{ph}$.

2. Equivalent spacing D_{eq}

$\sqrt[3]{D_{12}D_{23}D_{31}}$, the geometric mean of the three phase spacings; the same quantity appears in the inductance formula of 1(b).

4. Notes and Remarks

- **r, not D_s:** for capacitance the charge sits on the surface of the conductor, so the **outside radius** r is used. For inductance the GMR ($D_s = 0.7788\,r$ for a solid wire) is used.
- **Inductance and capacitance together:** $L = 2 \times 10^{-7}\ln(D_{eq}/D_s)$ and $C_n = 2\pi\varepsilon_0/\ln(D_{eq}/r)$ have the same shape. A wider spacing raises L and lowers C.
- **Effect of the earth:** the ground acts as image charges below the line and raises C slightly; it is neglected here, as in the textbook derivation.
- The same derivation is asked in the 2021-22 paper (5(d)), as extra reading.

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1.(d) A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line. Marks: 10

1. Theory

1. What is mutual inductance? (Core Idea)

When current flows in one circuit, part of its magnetic flux may also pass through a second circuit nearby. If the current is AC, this flux changes all the time and induces a voltage in the second circuit, although the two are not connected. **Mutual inductance** M measures this coupling:

$$M = \frac{\psi_{21}}{I_1}, \qquad E_2 = \omega M I_1 = 2\pi f M I_1$$

where ψ_{21} is the flux linkage of circuit 2 due to the current I_1 in circuit 1, and E_2 the voltage induced in circuit 2.
Power lines and telephone lines often run along the same road for many kilometres. A power line carries hundreds of amperes, so even a small M can induce a voltage in the telephone line. It is heard as a hum and can be dangerous.

2. Which flux links the telephone loop?

The telephone line is a loop made of two wires, T_1 and T_2. From 1(b), a straight wire carrying I makes rings of flux, and the rings with radius between D_{near} and D_{far} carry

$$\psi = 2 \times 10^{-7}\, I \ln \frac{D_{far}}{D_{near}} \quad \text{Wb-T/m}$$

A ring of a power wire links the telephone loop only if it passes **between** T_1 and T_2: its radius must be between the wire's distance to T_1 and its distance to T_2. Smaller rings go round neither telephone wire; larger rings go round both, and their effects in the two wires cancel.

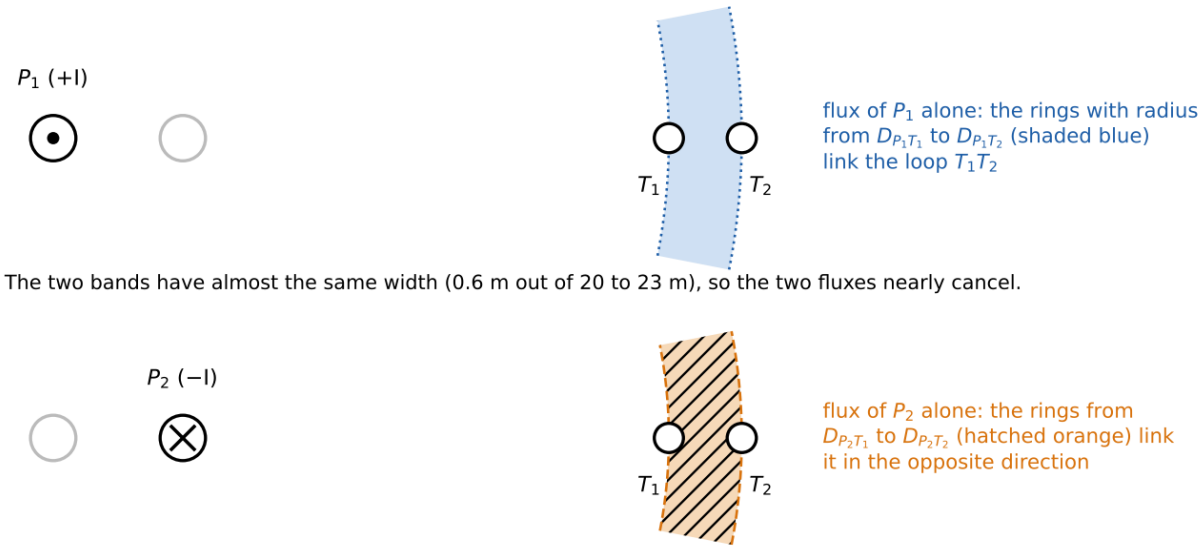


Figure 7: In one plane: the rings of P1 (shaded blue) and of P2 (hatched orange) that pass between T1 and T2; they link the loop in opposite directions (not to scale)

Reading the figure. Top row: P_1 alone, carrying $+I$. Its rings with radius from $D_{P_1T_1}$ to $D_{P_1T_2}$ (the shaded blue band) pass between T_1 and T_2. Bottom row: P_2 alone, carrying the return current $-I$. Its band (hatched orange) also passes between T_1 and T_2, but its current is opposite, so its flux through

the loop is opposite. Each band is about 0.6 m wide at a distance of 20 to 23 m, so the two fluxes are almost equal and nearly cancel.

3. The general rule

Adding the two wires, with the signs of their currents:

$$\psi = 2\times10^{-7}\, I\left[\ln\frac{D_{P_1T_2}}{D_{P_1T_1}} - \ln\frac{D_{P_2T_2}}{D_{P_2T_1}}\right] = 2\times10^{-7}\, I\ln\frac{D_{P_1T_2}\,D_{P_2T_1}}{D_{P_1T_1}\,D_{P_2T_2}}$$

so the mutual inductance per metre is

$$M = 2\times10^{-7}\ln\frac{D_{P_1T_2}\,D_{P_2T_1}}{D_{P_1T_1}\,D_{P_2T_2}}\ \text{H/m}$$

where $D_{P_1T_2}$ is the distance from P_1 to T_2, and so on. Its sign only gives the direction of the induced voltage.

4. How the interference is reduced

- Keep the lines far apart.
- Transpose the power line or the telephone line, so that the voltages induced in different sections cancel.
- Place the telephone wires where $D_{P_1T_2}D_{P_2T_1} = D_{P_1T_1}D_{P_2T_2}$ (for example both on the centre line between the power wires, one above the other); then $M = 0$.

2. Answer

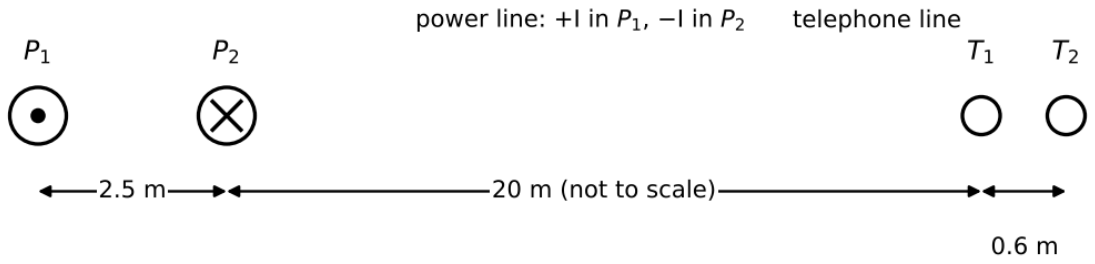


Figure 8: Power line P1, P2 and telephone line T1, T2 in one horizontal plane (reused from the 2021-22 paper)

Given:

- Single-phase power line, $f = 50$ Hz, $I = 150$ A, conductors $P_1P_2 = 2.5$ m apart
- Telephone conductors $T_1T_2 = 0.6$ m apart (solid copper; their size is not needed for M)
- All four conductors in one horizontal plane; nearest conductors P_2 and T_1 are 20 m apart

Relationships:

$$M = 2\times10^{-7}\ln\frac{D_{P_1T_2}\,D_{P_2T_1}}{D_{P_1T_1}\,D_{P_2T_2}}\ \text{H/m}, \qquad E = 2\pi f M I$$

Step 1: Distances

Put P_1 at 0. Then P_2 is at 2.5 m, T_1 at $2.5 + 20 = 22.5$ m and T_2 at $22.5 + 0.6 = 23.1$ m:

$$D_{P_1T_1} = 22.5\text{ m}, \quad D_{P_1T_2} = 23.1\text{ m}, \quad D_{P_2T_1} = 20\text{ m}, \quad D_{P_2T_2} = 20.6\text{ m}$$

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Step 2: Mutual inductance

$$\frac{D_{P_1T_2}\,D_{P_2T_1}}{D_{P_1T_1}\,D_{P_2T_2}} = \frac{23.1\times 20}{22.5\times 20.6} = \frac{462}{463.5} = 0.996764$$

$$M = 2\times 10^{-7}\ln(0.996764) = 2\times 10^{-7}\times(-0.0032415) = -6.483\times 10^{-10}\ \text{H/m}$$

The minus sign only fixes the direction of the induced voltage. Per kilometre ($\times$ 1000):

$$|M| = 6.483\times 10^{-7}\ \text{H/km}$$

$$\boxed{M = 0.648\ \mu\text{H per km}}$$

Step 3: Voltage induced in the telephone line

$$E = 2\pi f|M|I = 2\pi\times 50\times 6.483\times 10^{-7}\times 150 = 0.03055\ \text{V/km}$$

$$\boxed{E = 30.6\ \text{mV per km}}$$

The voltage is small because the telephone line is far away (20 m) compared with its own width (0.6 m), so the fluxes of P_1 and P_2 through it nearly cancel.

Final Answer: $M = 0.648\ \mu$H/km; induced voltage 30.6 mV per km.

3. Keyword Explanation

1. Mutual inductance

Flux linkage of one circuit per ampere of current in another circuit (H). It gives the voltage induced in the second circuit, $E = \omega M I$.

2. Electromagnetic interference

The unwanted voltage induced in communication lines by nearby power lines.

4. Notes and Remarks

- **The two contributions separately:** P_1 alone gives $2\times 10^{-7}\ln(23.1/22.5) = 5.264\times 10^{-9}$ H/m and P_2 alone $2\times 10^{-7}\ln(20.6/20) = 5.912\times 10^{-9}$ H/m, in opposite directions. The net M is their small difference, 6.48×10^{-10} H/m: about one eighth of either part.
- **Electrostatic induction** also exists (through the capacitance between the lines); the question asks only for the magnetic effect.
- The same question is solved in the 2021-22 paper as 9(b), and the case of a telephone line directly below the power line as 1(d), as extra reading.

5. Final Concise Answer

(a) Inductance is the flux linkage per ampere, $L = \psi/I = N\phi/I$ henry; the induced voltage is $e = L\,di/dt$.

(b) $D_s = 0.0466$ ft $= 0.014204$ m; $D_s^b = \sqrt{0.014204\times 0.45} = 0.07995$ m; $D_{eq} = \sqrt[3]{8\times 8\times 16} = 10.079$ m; $L = 0.9674$ mH/km; at 60 Hz $X_L = 0.365\ \Omega$/km $= 0.587\ \Omega$/mi per phase (50 Hz: 0.304 Ω/km).

(c) Averaging V_{ab} and V_{ac} over the three transposition sections with $q_a+q_b+q_c = 0$ and $V_{ab}+V_{ac} = 3V_{an}$ gives $C_n = 2\pi\varepsilon_0/\ln(D_{eq}/r)$ F/m per phase, $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$.

(d) $D_{P_1T_1} = 22.5$, $D_{P_1T_2} = 23.1$, $D_{P_2T_1} = 20$, $D_{P_2T_2} = 20.6$ m; $|M| = 2\times 10^{-7}|\ln(462/463.5)| = 0.648\ \mu$H/km; $E = 30.6$ mV/km.

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2. (a) Define Corona Effect. Marks: 1
   (b) Define Transposed Transmission line. Explain the reason behind Transpose. Marks: 4
   (c) Derive the expression of Inductance of composite conductor lines. Marks: 10
   (d) A 33 kV single core cable has a conductor diameter of 1 cm and a sheath of inside diameter 4 cm. Find the maximum and minimum stress in the insulation. Marks: 10

2.(a) Define Corona Effect. Marks: 1

1. Theory

1. Air is the only insulation of an overhead line (Core Idea)

An overhead line is made of bare conductors hung from towers. Between one conductor and the next, and between a conductor and the earth, there is only **air**. Air insulates well, but only up to a certain electric field.

2. The field is strongest at the conductor surface

A conductor at a high voltage makes an **electric field** (potential gradient g, in kV/cm) in the air around it. The field lines start on the conductor and spread outwards, so they are crowded near the surface and spread out further away. For two conductors of radius r and spacing d, with voltage V from each conductor to neutral, the field at distance x from a conductor's centre is

$$g = \frac{V}{x \ln(d/r)}$$

It is largest at the surface ($x = r$).

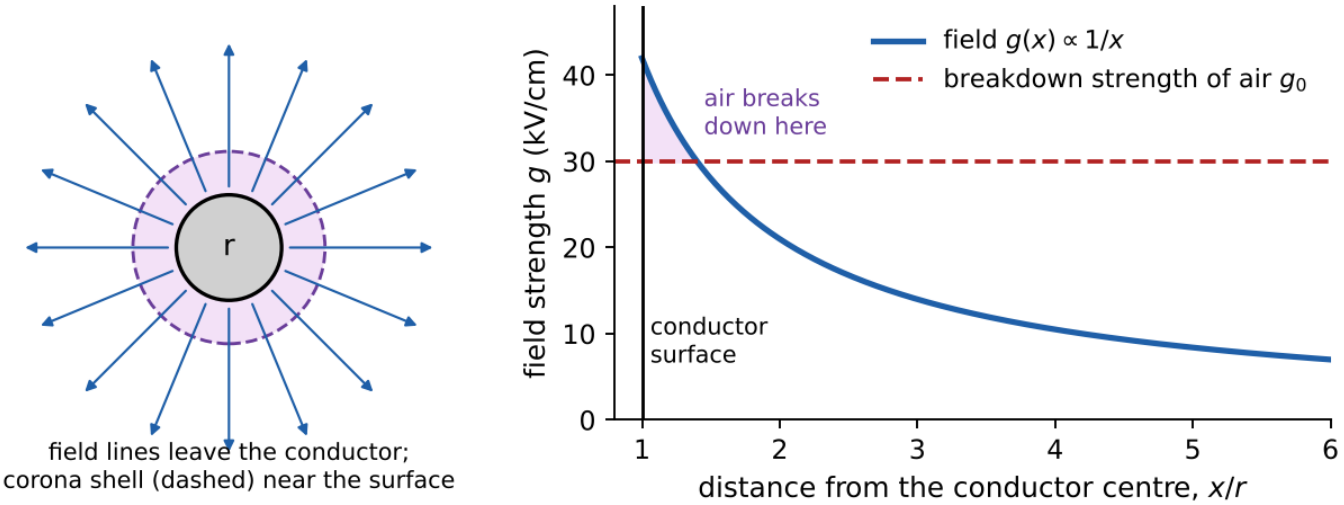


Figure 9: Left: field lines leave a conductor; the dashed ring is the thin shell of air near the surface. Right: the field falls as 1/x (solid blue); air breaks down only where it is above the dashed red line g0 (reused from the 2021-22 paper)

Reading the figure. The solid blue curve (right) is the field strength: high at the conductor surface ($x/r = 1$), falling quickly with distance. The dashed red line is the strength air can stand, g_0. Only the small shaded part, very close to the conductor, is above it.

3. Ionisation of the air

Air always contains a few free electrons. A strong field accelerates them so much that they knock more electrons out of air molecules, and the number grows like an avalanche: the air becomes conducting (**ionised**). This happens above the **dielectric strength of air**, $g_0 = 30$ kV/cm peak (21.2 kV/cm r.m.s.) at 76 cm of mercury and 25 °C. Because the field drops quickly with distance, only a thin shell of air round the conductor ionises; the breakdown does not reach the next conductor.

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2. Answer

Corona effect is the phenomenon of a **violet glow, a hissing sound, the production of ozone and a loss of power** around the conductors of an overhead line when the electric field at the conductor surface exceeds the dielectric strength of air (about 30 kV/cm peak): the air close to the conductor ionises and becomes partly conducting. It starts at the **critical disruptive voltage** $V_c = m_0 g_0 \delta r \ln(d/r)$ (see 4(a)).

3. Keyword Explanation

1. Ionisation

Free electrons, accelerated by a strong field, knock electrons out of neutral molecules, so the gas becomes conducting.

2. Dielectric strength of air (g_0)

The largest field air can stand without breaking down: 30 kV/cm peak at normal pressure and temperature.

4. Notes and Remarks

- **Effects of corona:** power loss, radio interference, ozone (which corrodes the conductor), and the nitric acid it can form in damp air. A small advantage: corona raises the effective radius of the conductor and so reduces the stress on surges.
- **Reduced by** a larger conductor diameter, wider spacing or bundled conductors (1(b)).

2.(b) Define Transposed Transmission line. Explain the reason behind Transpose. Marks: 4

1. Theory

1. Each phase sees its neighbours (Core Idea)

The inductance and capacitance of one phase depend on the distances to the other two phases (see 1(b) and 1(c)). The flux linkage of phase a, for example, is

$$\psi_a = 2 \times 10^{-7} \left[I_a \ln \frac{1}{D_s} + I_b \ln \frac{1}{D_{ab}} + I_c \ln \frac{1}{D_{ac}} \right]$$

2. Unequal distances give unequal phases

If the three conductors are at the corners of an equilateral triangle, all distances are equal and the three phases are alike. But most lines carry their conductors side by side on a cross-arm (flat spacing D, D, $2D$) or one above another. The middle conductor is then closer to both others than the outer conductors are. So:

- each phase has a different inductance and capacitance;
- the voltage drops of the three phases differ, so the receiving-end voltages are **unbalanced**, even with a balanced load;
- the line's magnetic and electric fields do not cancel at a distance, so it **induces** voltages in nearby telephone lines (1(d)).

2. Answer

Transposed transmission line: a three-phase line in which the conductors **exchange their positions at regular intervals** (at transposition towers), so that over one complete transposition cycle each conductor occupies each of the three positions for **one third of the length**.

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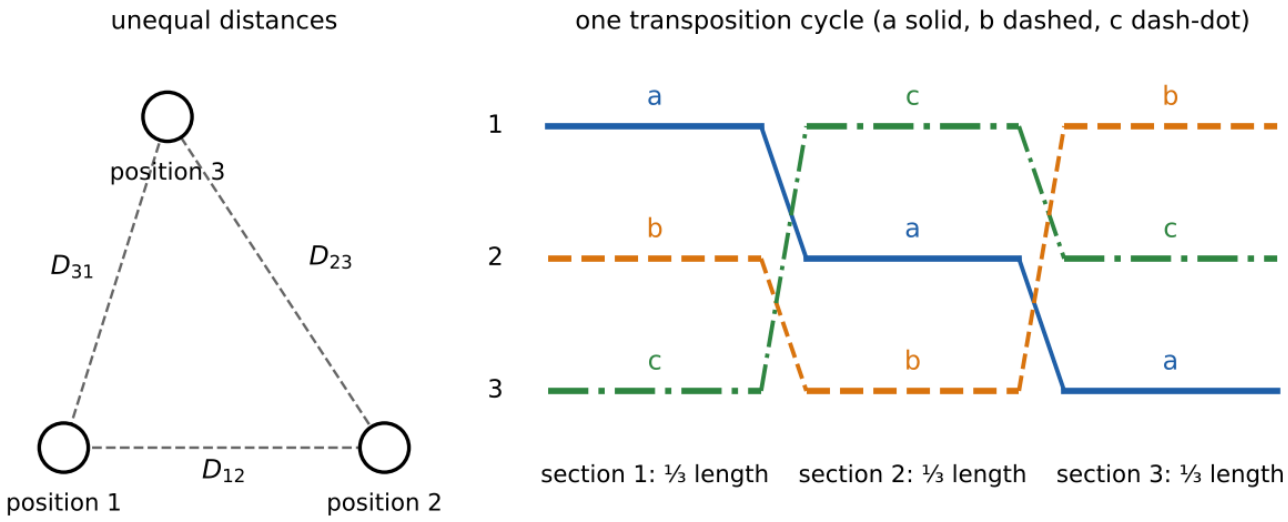


Figure 10: One transposition cycle: conductor a (solid blue), b (dashed orange) and c (dash-dot green) each take every position for one third of the length (reused from the 2021-22 paper)

Reason behind transposition:

- **Equal line constants.** When the spacing is unsymmetrical, each phase sees different distances to the others, so the phases have different inductances and capacitances. Transposition makes every conductor see the same surroundings on average, so the three phases get the same inductance $L = 2 \times 10^{-7} \ln(D_{eq}/D_s)$ and capacitance, with $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$.
- **Balanced voltages.** With equal constants, the voltage drops in the three phases are equal, so the receiving-end voltages stay balanced.
- **Less interference.** The voltages induced by the line in a nearby telephone or communication line in the three sections cancel, so the interference is reduced.
- **No circulating currents** are set up by the unbalance between phases.

3. Keyword Explanation

1. Transposition cycle

The length of line over which each conductor has taken each position once; the cycle is repeated along the line.

2. Unsymmetrical spacing

An arrangement in which the three distances between phases are not all equal.

4. Notes and Remarks

- **In practice** transposition is done at a few towers (or at substations) because each transposition tower costs more; a line may have one or two cycles over its whole length.
- **Untransposed lines** are common for short lines, where the unbalance is small; the calculation then uses the average values as an approximation.

2.(c) Derive the expression of Inductance of composite conductor lines. Marks: 10

1. Theory

1. What is a composite conductor? (Core Idea)

A real conductor is seldom one solid wire. It is made of many thin wires (**strands**) twisted together, so it bends easily. A conductor made of several parallel strands, all carrying current in the same direction, is a **composite conductor**. A bundled conductor (1(b)) is also a composite conductor.

We find its inductance by treating each strand as a small round wire and using two results of 1(b):

- a round wire of GMR r' carrying I has, up to a distance D, the flux linkage $2\times10^{-7}\,I\ln(D/r')$, with $r'=0.7788\,r$;
- when the currents of a group of wires add up to zero, the flux linkage of one wire is $2\times10^{-7}\sum_k I_k\ln(1/D_k)$, where D_k is its distance to wire k (and r' for itself).

2. Two new words: GMD and GMR

- **Mutual GMD (geometric mean distance), D_m:** take the distance from every strand of one conductor to every strand of the other, multiply them all, and take the root of the same order as the number of distances.
- **Self GMD (GMR), D_s:** do the same with the distances between every pair of strands **of the same conductor**, counting each strand's distance to itself as its own r'.

The GMD says how far the two conductors are from each other on average; the GMR says how "spread out" one conductor is. A more spread-out conductor has a larger GMR and a smaller inductance.

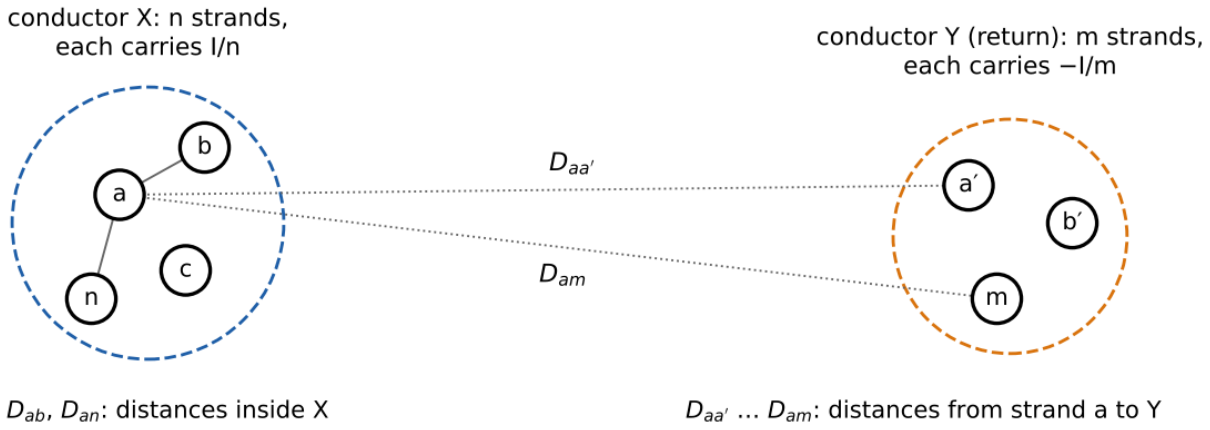


Figure 11: Conductor X with n strands, each carrying I/n, and the return conductor Y with m strands, each carrying −I/m (reused from the 2021-22 paper)

Reading the figure. Conductor X (left, n strands) carries the current I, split equally between its strands; conductor Y (right, m strands) carries it back. D_{ab} and D_{an} are distances between strands of X; $D_{aa'}$ to D_{am} are distances from strand a to the strands of Y.

2. Answer

Consider a single-phase line made of two composite conductors (figure above). Conductor X has n identical parallel strands, each carrying I/n; the return conductor Y has m identical strands, each carrying $-I/m$. The currents add up to zero, so the flux linkage of any strand is $2\times10^{-7}\sum I_k\ln(1/D_k)$.

Flux linkage of strand a of X (its own current and the other $n-1$ strands of X, then the m strands of Y):

$$\psi_a = 2\times10^{-7}\left[\frac{I}{n}\left(\ln\frac{1}{r'_a}+\ln\frac{1}{D_{ab}}+\cdots+\ln\frac{1}{D_{an}}\right)-\frac{I}{m}\left(\ln\frac{1}{D_{aa'}}+\ln\frac{1}{D_{ab'}}+\cdots+\ln\frac{1}{D_{am}}\right)\right]$$

Combining the logarithms:

$$\psi_a = 2\times10^{-7}\,I\ln\frac{\sqrt[m]{D_{aa'}D_{ab'}\cdots D_{am}}}{\sqrt[n]{r'_a D_{ab}\cdots D_{an}}}\ \text{Wb-T/m}$$

Inductance of strand a (it carries I/n):

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$$L_a = \frac{\psi_a}{I/n} = 2n \times 10^{-7} \ln \frac{\sqrt[m]{D_{aa'} D_{ab'} \cdots D_{am}}}{\sqrt[n]{r'_a D_{ab} \cdots D_{an}}} \ \text{H/m}$$

Inductance of strand b, in the same way:

$$L_b = 2n \times 10^{-7} \ln \frac{\sqrt[m]{D_{ba'} D_{bb'} \cdots D_{bm}}}{\sqrt[n]{D_{ba} r'_b \cdots D_{bn}}} \ \text{H/m}$$

and so on for $L_c, \ldots, L_n$.

Inductance of conductor X. The average inductance of one strand is

$$L_{av} = \frac{L_a + L_b + \cdots + L_n}{n}$$

The n strands are in parallel, all with (nearly) this inductance, so

$$L_X = \frac{L_{av}}{n} = \frac{L_a + L_b + \cdots + L_n}{n^2}$$

Putting in the expressions for $L_a, \ldots, L_n$ (each has the factor $2n \times 10^{-7}$, and the sum of n logarithms divided by n becomes one logarithm of an n-th root):

$$L_X = 2 \times 10^{-7} \ln \frac{\sqrt[mn]{(D_{aa'} D_{ab'} \cdots D_{am})(D_{ba'} D_{bb'} \cdots D_{bm}) \cdots (D_{na'} \cdots D_{nm})}}{\sqrt[n^2]{(r'_a D_{ab} \cdots D_{an})(D_{ba} r'_b \cdots D_{bn}) \cdots (D_{na} D_{nb} \cdots r'_n)}}$$

$$\boxed{L_X = 2 \times 10^{-7} \ln \frac{D_m}{D_{sX}} \ \text{H/m}}$$

where

- $D_m = \sqrt[mn]{(D_{aa'} D_{ab'} \cdots D_{am}) \cdots (D_{na'} \cdots D_{nm})}$ is the **mutual GMD** between X and Y (mn distances), and
- $D_{sX} = \sqrt[n^2]{(D_{aa} D_{ab} \cdots D_{an}) \cdots (D_{na} \cdots D_{nn})}$, with $D_{aa} = r'_a$, $D_{bb} = r'_b$, …, is the **self GMD (GMR)** of X (n^2 distances).

Inductance of Y, in the same way: $L_Y = 2 \times 10^{-7} \ln(D_m / D_{sY})$.

Loop inductance of the line:

$$\boxed{L = L_X + L_Y = 2 \times 10^{-7} \ln \frac{D_m^2}{D_{sX} D_{sY}} \ \text{H/m}}$$

Check: for two solid wires ($n = m = 1$) a distance d apart, $D_m = d$ and $D_s = r'$, so $L = 4 \times 10^{-7} \ln(d/r')$, the familiar single-phase result.

3. Keyword Explanation

1. Mutual GMD

The mn-th root of the product of all distances between the strands of one conductor and the strands of the other.

2. Self GMD (GMR)

The n^2-th root of the product of all distances between the strands of one conductor, each strand's distance to itself taken as $r' = 0.7788\, r$.

4. Notes and Remarks

- **Why GMD and GMR are so useful:** the inductance of any arrangement keeps the same simple form, $2 \times 10^{-7} \ln(\text{GMD}/\text{GMR})$; only the two distances change. For a three-phase line the

GMD is the equivalent spacing D_{eq}.

- **Example of a GMR**

  the bundle of 1(b) is a composite conductor with $n = 2$: $D_s^b = \sqrt[4]{D_s \cdot d \cdot d \cdot D_s} = \sqrt{D_s\, d}$.

- **Stranded conductors:** the GMR of a 7-strand conductor is $0.726\,R$ and of a 19-strand conductor $0.758\,R$, where R is the overall radius.
- The same derivation is asked in the 2021-22 paper (4(c)), as extra reading.

2.(d) A 33 kV single core cable has a conductor diameter of 1 cm and a sheath of inside diameter 4 cm. Find the maximum and minimum stress in the insulation. Marks: 10

1. Theory

1. The single-core cable (Core Idea)

An underground cable has a round copper or aluminium **core** at the centre, a layer of **insulation** (impregnated paper, PVC or XLPE) around it, and an earthed metal **sheath** outside. The core is at the line voltage and the sheath at earth, so the whole voltage V is held by the insulation between them. The cable is a **capacitor** shaped like a tube.

2. Electric stress

The **electric stress** (potential gradient) g at a point of the insulation is the electric field there: the voltage drop per centimetre, in kV/cm. If the stress anywhere goes above the **dielectric strength** of the insulation, the insulation breaks down. So we must know where the stress is largest and how large it is.

3. How the stress varies across the insulation

Let the core carry a charge $+Q$ per metre (the sheath $-Q$). The field points straight out from the core. By Gauss's law, the charge spreads its flux over a cylinder of radius x (surface $2\pi x$ per metre), so

$$g = \frac{Q}{2\pi\varepsilon_0\varepsilon_r\, x}$$

Adding the field across the insulation from the core (radius $d/2$) to the sheath (radius $D/2$) gives the voltage, $V = \dfrac{Q}{2\pi\varepsilon_0\varepsilon_r} \ln\dfrac{D}{d}$. Removing Q:

$$g = \frac{V}{x\,\ln(D/d)}$$

(The full derivation is 3(c).) The stress falls as $1/x$:

- **largest** at the core surface, $x = d/2$: $g_{max} = \dfrac{2V}{d\,\ln(D/d)}$
- **smallest** at the sheath, $x = D/2$: $g_{min} = \dfrac{2V}{D\,\ln(D/d)}$

and $g_{max}/g_{min} = D/d$. The stress does not depend on the permittivity: only on the voltage and the two diameters.

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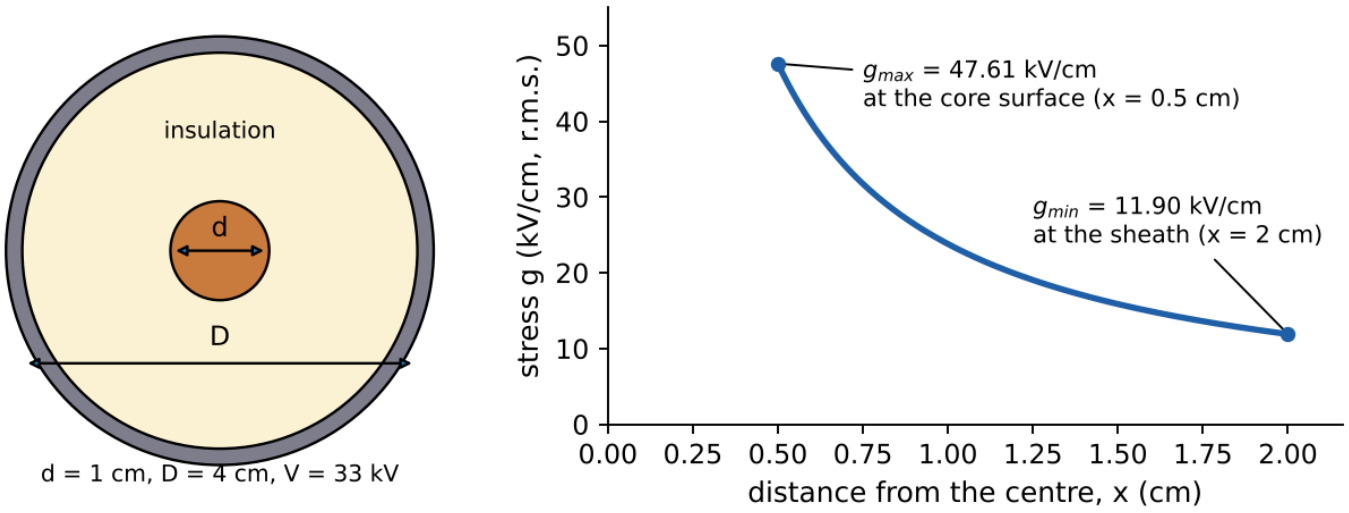


Figure 12: Left: section of the cable of this question. Right: the stress across its insulation, 47.61 kV/cm at the core falling to 11.90 kV/cm at the sheath

Reading the figure. On the left, the core (diameter $d = 1$ cm) sits inside the insulation and the sheath (inside diameter $D = 4$ cm). On the right, the solid blue curve is the stress: 47.61 kV/cm at the core surface ($x = 0.5$ cm), falling as $1/x$ to 11.90 kV/cm at the sheath ($x = 2$ cm). The insulation near the core works four times harder than the insulation near the sheath.

2. Answer

Given:

- Voltage between core and sheath $V = 33$ kV (r.m.s.)
- Conductor diameter $d = 1$ cm
- Inside diameter of the sheath $D = 4$ cm

Assumption: following V.K. Mehta, the 33 kV is taken as the voltage between the core and the earthed sheath. (If it were the line voltage of a three-phase system, the core-to-sheath voltage would be $33/\sqrt{3} = 19.05$ kV; see Notes.)

Relationships:

$$g_{max} = \frac{2V}{d\,\ln(D/d)}, \qquad g_{min} = \frac{2V}{D\,\ln(D/d)}$$

Step 1: The logarithm

$$\frac{D}{d} = \frac{4}{1} = 4, \qquad \ln 4 = 1.3863$$

Step 2: Maximum stress (at the conductor surface)

$$g_{max} = \frac{2 \times 33}{1 \times 1.3863} = 47.61 \text{ kV/cm}$$

$$\boxed{g_{max} = 47.61 \text{ kV/cm r.m.s.}}$$

Step 3: Minimum stress (at the sheath)

$$g_{min} = \frac{2 \times 33}{4 \times 1.3863} = 11.90 \text{ kV/cm}$$

$$\boxed{g_{min} = 11.90 \text{ kV/cm r.m.s.}}$$

Check: $g_{max}/g_{min} = 47.61/11.90 = 4 = D/d$.

Final Answer: maximum stress 47.61 kV/cm r.m.s. (at the core), minimum stress 11.90 kV/cm r.m.s.

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(at the sheath).

3. Keyword Explanation

1. Electric (dielectric) stress

The electric field in the insulation, in kV/cm; the insulation fails where it exceeds the dielectric strength.

2. Dielectric strength

The largest stress an insulating material can stand without breaking down.

4. Notes and Remarks

- **Peak values:** multiplying by $\sqrt{2}$ gives $g_{max} = 67.33$ kV/cm and $g_{min} = 16.83$ kV/cm peak.
- **If 33 kV is the line voltage** of a three-phase system, $V = 19.05$ kV and both stresses are $\sqrt{3}$ times smaller: $g_{max} = 27.49$ kV/cm, $g_{min} = 6.87$ kV/cm r.m.s.
- **Uneven stress** is why cables are **graded** (layers of different permittivity, or intersheaths), so that the insulation is used more evenly.

5. Final Concise Answer

(a) Corona is the violet glow, hissing noise, ozone and power loss round a line conductor when the field at its surface exceeds the strength of air (30 kV/cm peak), so the air next to it ionises.

(b) A transposed line exchanges conductor positions at intervals so that each conductor occupies each position for one third of the length. Reason: with unsymmetrical spacing the phases have unequal L and C; transposition equalises them, keeps the voltages balanced and cancels the interference in nearby telephone lines.

(c) For conductor X (n strands) and its return Y (m strands): $L_X = 2 \times 10^{-7} \ln(D_m/D_{sX})$ H/m, with D_m the mn-th root of all distances between X and Y and D_{sX} the n^2-th root of all distances within X ($D_{aa} = r'$); loop $L = 2 \times 10^{-7} \ln[D_m^2/(D_{sX} D_{sY})]$.

(d) $\ln(D/d) = \ln 4 = 1.3863$; $g_{max} = 2V/(d \ln 4) = 47.61$ kV/cm r.m.s.; $g_{min} = 2V/(D \ln 4) = 11.90$ kV/cm r.m.s.

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3. (a) Define Skin Effect. Marks: 1
   (b) Describe different FACTS devices and applications. Marks: 4
   (c) Derive the expression of dielectric stress in a single core cable. Marks: 10
   (d) The capacitances of a 3-phase belted cable are $12 \cdot 6$ μF between the three cores bunched together and the lead sheath and $7 \cdot 4$ μF between one core and the other two connected to sheath. Find the charging current drawn by the cable when connected to 66 kV, 50 Hz supply. Marks: 10

3.(a) Define Skin Effect. Marks: 1

1. Theory

1. Direct current uses the whole conductor (Core Idea)

With **direct current** (DC), the current spreads evenly over the whole cross-section of a conductor. The resistance is then $R = \rho l/A$, with A the full area.

2. Alternating current is pushed to the surface

Think of the conductor as many thin parallel filaments. With **alternating current** (AC), each filament is surrounded by the magnetic flux of the current (1(a)). A filament at the **centre** is surrounded by all the flux, inside and outside the conductor; a filament at the **surface** is surrounded only by the flux outside. So the central filaments have more flux linkage, hence more inductance and more reactance. The AC current takes the easier path: it crowds towards the surface, and the centre carries less.

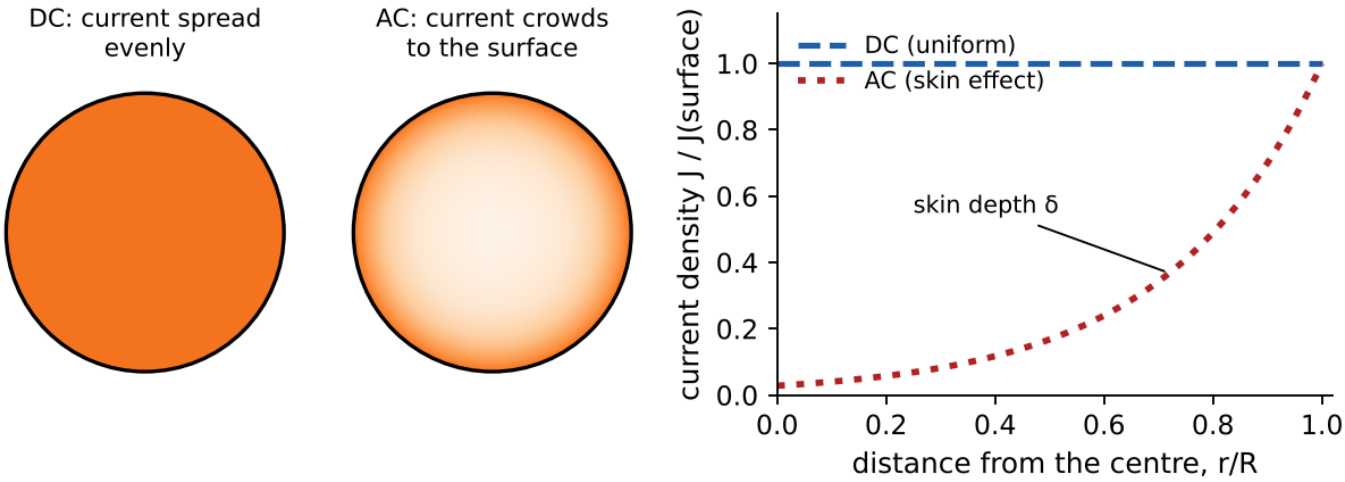


Figure 13: Current density across a round conductor: uniform for DC (left), crowded towards the surface for AC (middle); right, the density against the radius, DC dashed blue and AC dotted red

Reading the figure. On the left, the DC current density is the same everywhere (even colour). In the middle, with AC, the colour is strong only near the surface. On the right, the dashed blue line (DC) is flat; the dotted red curve (AC) is high at the surface ($r/R = 1$) and falls towards the centre. At one **skin depth** δ below the surface it has fallen to $1/e$ (37 %) of its surface value.

3. What it changes

Since the centre carries little current, the conductor behaves as if its area were smaller. Its **AC resistance is higher** than its DC resistance, so the I^2R loss is higher. The effect grows with the frequency, the conductor diameter, the conductivity and the permeability (steel conductors show it strongly).

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2. Answer

Skin effect is the tendency of an **alternating current** to concentrate near the **surface** of a conductor instead of spreading evenly over its cross-section. The central filaments of the conductor are linked by more flux, so they have more inductive reactance than the outer ones, and the current density is highest at the surface. As a result, the effective area is reduced and the **AC resistance is greater than the DC resistance** ($R_{ac} > R_{dc}$). The effect increases with frequency, conductor diameter and permeability; it is absent with DC.

3. Keyword Explanation

1. Skin depth (δ)

The depth below the surface at which the AC current density has fallen to $1/e$ of its surface value; $\delta = 1/\sqrt{\pi f \mu \sigma}$ (about 9 mm for copper at 50 Hz).

4. Notes and Remarks

- **At power frequency** the skin effect is small for thin conductors but noticeable for large ones; this is one reason why large conductors are **stranded** or made **hollow**, and why ACSR puts the steel (which carries little current) at the centre.
- **Proximity effect** is a related effect: the current in one conductor is pushed to one side by the field of a nearby conductor.

3.(b) Describe different FACTS devices and applications. Marks: 4

1. Theory

1. What limits the power on a line? (Core Idea)

The real power sent over a line of series reactance X, between two ends at voltages $V_1\angle\delta$ and $V_2\angle 0$, is

$$P = \frac{V_1 V_2}{X} \sin\delta$$

So the power flow is set by three quantities: the **voltages** V_1, V_2, the **line reactance** X, and the **angle** δ between the two ends. In an ordinary AC network none of these can be changed quickly: the power simply divides between parallel paths according to their reactances, some lines get overloaded while others are under-used, and the line must be run well below its thermal limit to stay stable.

2. The FACTS idea

FACTS (flexible AC transmission systems) puts fast **power-electronic controllers** (thyristors, or turn-off devices such as IGBTs and GTOs) into the network, so that V, X or δ can be changed in a few milliseconds. The line can then carry more power, closer to its thermal limit, and the flow can be steered where it is wanted. Hingorani and Gyugyi group the controllers by how they are connected:

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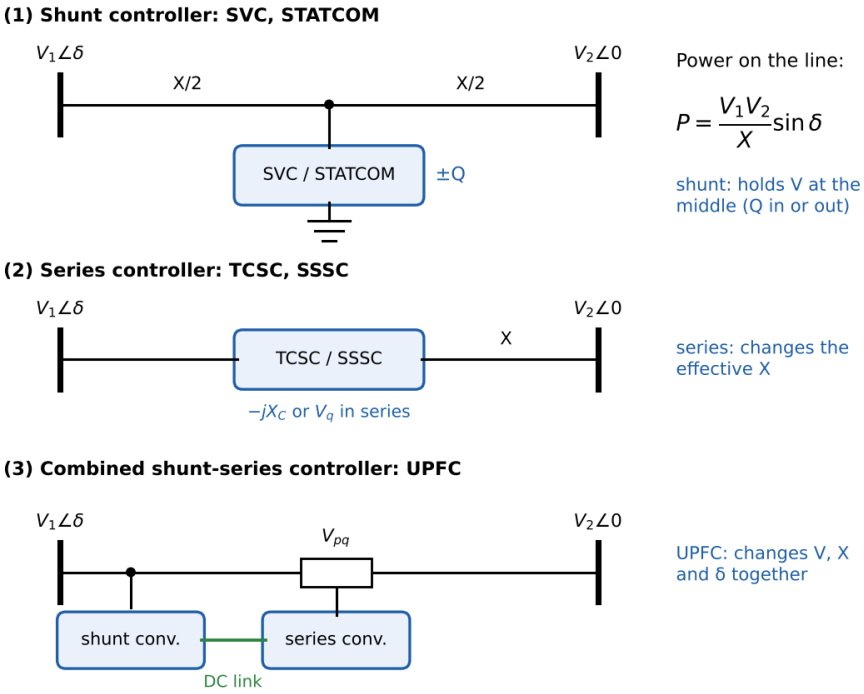


Figure 14: The three kinds of FACTS controller: shunt (SVC, STATCOM), series (TCSC, SSSC) and combined shunt-series (UPFC), with what each changes in P = V1V2 sin(delta)/X

Reading the figure. (1) A **shunt** controller is connected from a point of the line to earth; it injects or absorbs reactive power and so holds the voltage there. (2) A **series** controller is put in the line itself; it adds a capacitive reactance (or a voltage in series), which changes the effective X. (3) A **combined** controller (UPFC) has a shunt converter and a series converter joined by a DC link; it can change the voltage, the effective reactance and the angle together.

2. Answer

FACTS (flexible AC transmission systems) are power-electronic controllers that control the voltage, impedance and phase angle of an AC transmission line, to increase its power transfer capability and to control the power flow (Hingorani and Gyugyi).

Type	Devices	How it works	Main applications
Shunt	**SVC** (static VAR compensator: thyristor-controlled reactor + thyristor-switched capacitor); **STATCOM** (static synchronous compensator: a voltage-source converter)	Injects or absorbs reactive power at a bus	Voltage control and reactive power support, prevention of voltage collapse, damping of power swings, more transfer at the mid-point of long lines

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Type	Devices	How it works	Main applications
Series	**TCSC** (thyristor-controlled series capacitor); **SSSC** (static synchronous series compensator)	Adds a controllable capacitive reactance, or a voltage in quadrature with the current, in series with the line	Raises the power transfer (X reduced), controls the power flow in a chosen line, damps oscillations and sub-synchronous resonance, improves transient stability
Combined series-shunt	**UPFC** (unified power flow controller); TCPAR (thyristor-controlled phase-angle regulator)	A shunt and a series converter linked through a DC capacitor; inject a voltage of any phase in series	Controls the real and reactive power flow independently, regulates voltage, impedance and angle together
Combined series-series	**IPFC** (interline power flow controller)	Series converters in several lines sharing a DC link	Balances the power flow between lines

Table 2: Summary

Applications in short: more power over existing lines (close to their thermal limit), control of power flow (relieving overloaded lines and loop flows), voltage control, better transient and steady-state stability, damping of power oscillations, and faster recovery after faults.

3. Keyword Explanation

1. Voltage-source converter (VSC)

A power-electronic converter, using turn-off devices, that makes an AC voltage of controllable size and phase from a DC capacitor; the basis of STATCOM, SSSC and UPFC.

2. Reactive power compensation

Supplying or absorbing reactive power where it is needed, so that the voltage stays near its rated value.

4. Notes and Remarks

- **Thyristor-based** controllers (SVC, TCSC) act like variable reactors or capacitors; **converter-based** controllers (STATCOM, SSSC, UPFC) act like controllable voltage sources and keep their effect even at low voltage.
- **HVDC links** are also power-electronic and control power exactly, but they are not counted as FACTS controllers, which work on AC lines.

3.(c) Derive the expression of dielectric stress in a single core cable. Marks: 10

1. Theory

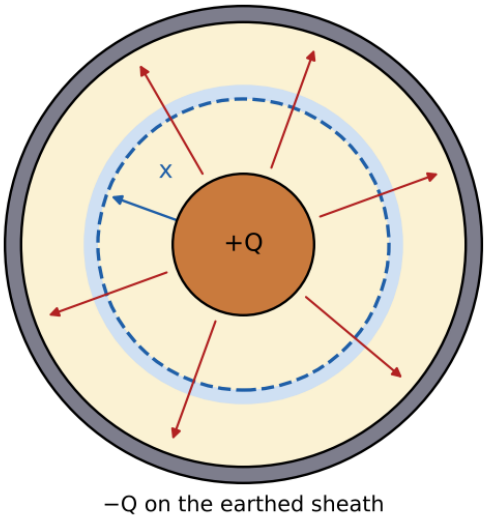
1. The cable as a cylindrical capacitor (Core Idea)

A single-core cable has a core of diameter d, insulation of relative permittivity ε_r, and an earthed sheath of inside diameter D (2(d)). With voltage V between core and sheath, a charge $+Q$ coulomb per metre sits on the core and $-Q$ on the inside of the sheath. Because everything is round, the electric field in the insulation points straight outwards from the centre, and has the same strength all round a circle of radius x.

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2. Two tools: Gauss's law and the voltage as an integral

- **Gauss's law:** the electric flux leaving a closed surface equals the charge inside it. The **electric flux density** D_x (C/m^2) through a cylinder of radius x and length 1 m (surface $2\pi x$) is therefore $D_x = Q/(2\pi x)$.
- **Field from flux density:** in a material of permittivity $\varepsilon = \varepsilon_0\varepsilon_r$, the field is $g = D_x/\varepsilon$.
- **Voltage:** the voltage across a thin shell of thickness dx is $g\,dx$; adding all the shells from the core to the sheath gives the whole voltage V.



core radius d/2 (brown), sheath radius D/2 (grey)

dashed blue: an imaginary cylinder of radius x, 1 m long; its surface is 2πx m²

shaded ring: a thin shell of thickness dx; the voltage across it is g dx

red arrows: the field points straight out from the core to the sheath

$$D_x = \frac{Q}{2\pi x}, \quad g_x = \frac{Q}{2\pi\varepsilon_0\varepsilon_r x}$$

Figure 15: Section of a single-core cable: +Q on the core, an imaginary cylinder of radius x (dashed blue), a thin shell dx (shaded) and the radial field (red arrows)

Reading the figure. The brown core carries $+Q$ per metre and the grey sheath $-Q$. The red arrows show the field pointing straight out. The dashed blue circle is the imaginary cylinder of radius x on which Gauss's law is applied; the thin shaded ring next to it is a shell of thickness dx, across which the voltage is $g\,dx$.

2. Answer

Set-up. Let the cable have core diameter d (radius $d/2$), sheath inside diameter D (radius $D/2$), insulation of relative permittivity ε_r, and voltage V between core and sheath. Let Q be the charge per metre on the core (figure above).

Step 1: Flux density at radius x. Consider a cylinder of radius x ($d/2 < x < D/2$) and 1 m length, coaxial with the core. Its curved surface is $2\pi x \times 1$ m^2. By Gauss's law all the flux Q passes through it, evenly:

$$D_x = \frac{Q}{2\pi x} \ \text{C/m}^2$$

Step 2: Electric stress at radius x.

$$g_x = \frac{D_x}{\varepsilon_0\varepsilon_r} = \frac{Q}{2\pi\varepsilon_0\varepsilon_r\, x} \ \text{V/m} \qquad (1)$$

Step 3: Voltage between core and sheath. The voltage across a shell of thickness dx at radius x is $g_x\,dx$. Adding all shells from the core to the sheath:

$$V = \int_{d/2}^{D/2} g_x\,dx = \int_{d/2}^{D/2} \frac{Q}{2\pi\varepsilon_0\varepsilon_r\, x}\,dx = \frac{Q}{2\pi\varepsilon_0\varepsilon_r}\ln\frac{D/2}{d/2} = \frac{Q}{2\pi\varepsilon_0\varepsilon_r}\ln\frac{D}{d} \qquad (2)$$

Step 4: Remove Q. From (2), $\dfrac{Q}{2\pi\varepsilon_0\varepsilon_r} = \dfrac{V}{\ln(D/d)}$. Putting this into (1):

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$$\boxed{g_x = \frac{V}{x\,\ln(D/d)} \text{ V/m}}$$

So the dielectric stress is **inversely proportional to the distance** x from the centre: it is not uniform across the insulation.

Step 5: Maximum and minimum stress.

- At the conductor surface, $x = d/2$ (the stress is largest):

$$\boxed{g_{max} = \frac{2V}{d\,\ln(D/d)}}$$

- At the inside of the sheath, $x = D/2$ (the stress is smallest):

$$\boxed{g_{min} = \frac{2V}{D\,\ln(D/d)}}$$

and $\dfrac{g_{max}}{g_{min}} = \dfrac{D}{d}$.

Step 6 (consequence): capacitance. From (2), $C = Q/V = \dfrac{2\pi\varepsilon_0\varepsilon_r}{\ln(D/d)}$ F/m.

Interpretation: the insulation next to the core carries the highest stress and must be strong enough for g_{max}; the outer layers are lightly stressed. The stress at the core is lowest (for a fixed V and D) when $D/d = e = 2.718$, which gives the most economical core size; 2(d) shows the stress profile for a real cable.

3. Keyword Explanation

1. Electric flux density (D_x)

The electric flux per square metre, C/m^2; around a long line charge it is $Q/(2\pi x)$.

2. Permittivity ($\varepsilon_0\varepsilon_r$)

How easily a material lets electric flux through; $\varepsilon_0 = 8.854\times10^{-12}$ F/m and ε_r is the relative permittivity of the insulation (3 to 4 for impregnated paper).

4. Notes and Remarks

- **Units:** with V in kV and x, d, D in cm, g comes out in kV/cm, the usual unit.
- **Most economical core:** for a given V and allowed g_{max}, the sheath diameter D is smallest when $\ln(D/d) = 1$, so $d = 2V/g_{max}$ and $D = e\,d$.
- **Grading** (capacitance grading with layers of different ε_r, or intersheath grading) makes the stress more uniform, so that thinner insulation can be used.
- The same derivation appears inside 2(c) of the 2021-22 paper, as extra reading.

3.(d) The capacitances of a 3-phase belted cable are $12 \cdot 6$ μF between the three cores bunched together and the lead sheath and $7 \cdot 4$ μF between one core and the other two connected to sheath. Find the charging current drawn by the cable when connected to 66 kV, 50 Hz supply. Marks: 10

1. Theory

1. A three-core belted cable (Core Idea)

For three-phase supply up to about 22 kV, the three conductors (cores) are often put in *one* cable. Each core has its own paper insulation; the three are laid together with jute fillers, wrapped in a common

paper **belt** insulation, and enclosed in one lead **sheath**. This is a **belted cable**.

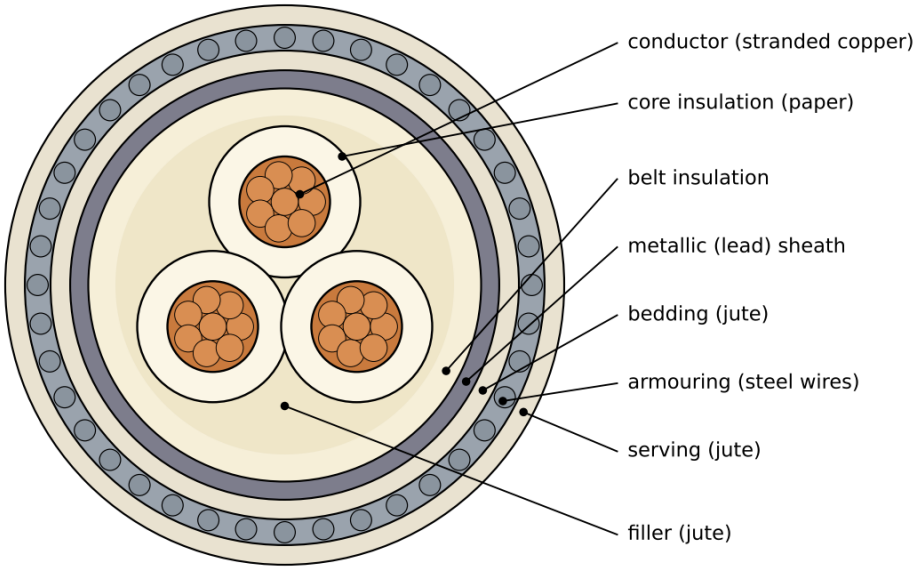
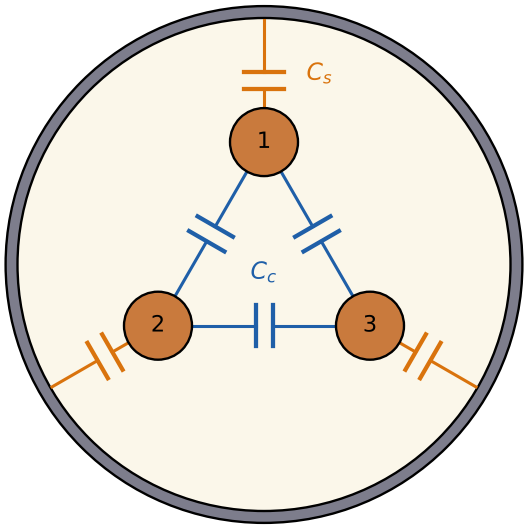


Figure 16: Construction of a three-core belted cable: three stranded conductors with their insulation, belt insulation, lead sheath, bedding, armouring and serving (reused from the 2021-22 paper)

2. Its capacitances

Every two metal surfaces separated by insulation form a capacitor. In a belted cable there are two kinds:

- C_c: between **each pair of cores** (three of them, in a delta);
- C_s: between **each core and the sheath** (three of them).



each pair of cores: C_c (blue)

each core to the sheath: C_s (orange)

The delta of three C_c is the same as a star of three $3C_c$ to a neutral N at the sheath's potential, so each core sees, to neutral,

$$C_N = C_s + 3C_c$$

(delta to star for capacitors: $C_Y = 3C_\Delta$)

Figure 17: Capacitances of a belted cable: Cc between each pair of cores (blue) and Cs from each core to the sheath (orange); per phase they add to CN = Cs + 3Cc

Reading the figure. The three cores 1, 2, 3 sit inside the sheath. Blue capacitors join every pair of cores (C_c); orange capacitors join each core to the sheath (C_s).

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3. Capacitance per phase (to neutral)

The charging current of a three-phase circuit is found per phase, from the capacitance between one core and the **neutral**. The three C_c form a **delta**. A delta of three equal capacitors C_c behaves like a **star** of three capacitors $3C_c$ whose star point is the neutral. With balanced voltages the sheath is at the neutral potential, so for each core the $3C_c$ to neutral is in parallel with its C_s to the sheath:

$$C_N = C_s + 3C_c$$

The charging current per phase is then $I_c = \omega C_N V_{ph}$, where $V_{ph} = V_L/\sqrt{3}$.

4. Measuring C_s and C_c

C_s and C_c cannot be measured directly, but two simple measurements give them:

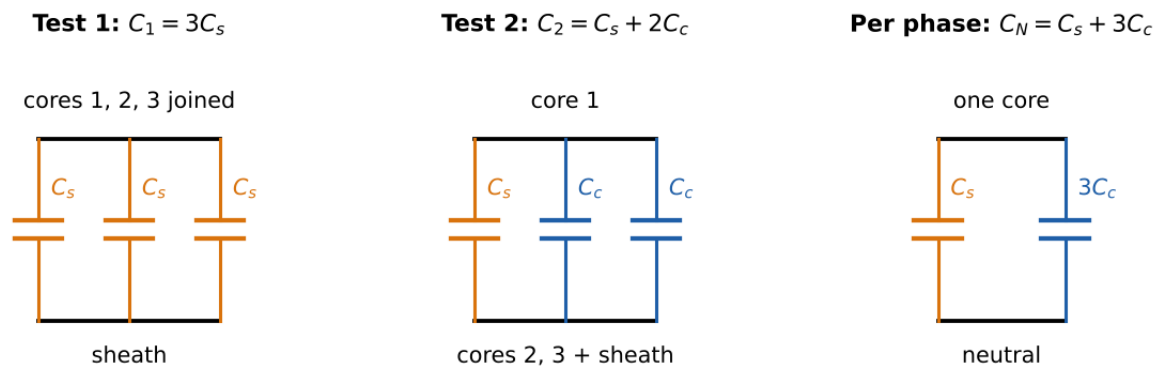


Figure 18: The two measurements and the per-phase capacitance drawn as parallel capacitors: test 1 gives 3Cs, test 2 gives Cs + 2Cc, and per phase CN = Cs + 3Cc

Reading the figure.

- **Test 1 (left):** the three cores are joined together and the capacitance is measured between them and the sheath. The C_c now have both ends joined, so they carry no charge; only the three C_s remain, in parallel: $C_1 = 3C_s$.
- **Test 2 (middle):** one core is measured against the other two cores joined to the sheath. That core sees its own C_s and the two C_c to the other cores, all in parallel: $C_2 = C_s + 2C_c$.
- **Per phase (right):** $C_N = C_s + 3C_c$.

2. Answer

Given:

- Three cores bunched together to the sheath: $C_1 = 3C_s = 12.6\ \mu$F
- One core to the other two connected to the sheath: $C_2 = C_s + 2C_c = 7.4\ \mu$F
- Supply $V_L = 66$ kV (line voltage), $f = 50$ Hz

Relationships:

$$C_1 = 3C_s, \qquad C_2 = C_s + 2C_c, \qquad C_N = C_s + 3C_c, \qquad I_c = 2\pi f\, C_N\, V_{ph}$$

Step 1: Core-to-sheath capacitance

$$C_s = \frac{C_1}{3} = \frac{12.6}{3} = 4.2\ \mu\text{F}$$

Step 2: Core-to-core capacitance

$$C_c = \frac{C_2 - C_s}{2} = \frac{7.4 - 4.2}{2} = 1.6\ \mu\text{F}$$

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Step 3: Capacitance per phase (core to neutral)

$$C_N = C_s + 3C_c = 4.2 + 3 \times 1.6 = 9.0 \ \mu\text{F}$$

$$\boxed{C_N = 9.0 \ \mu\text{F per phase}}$$

Step 4: Charging current

The phase voltage is

$$V_{ph} = \frac{66\,000}{\sqrt{3}} = 38\,105 \text{ V}$$

$$I_c = 2\pi f \, C_N \, V_{ph} = 2\pi \times 50 \times 9.0 \times 10^{-6} \times 38\,105 = 107.7 \text{ A}$$

$$\boxed{I_c = 107.7 \text{ A}}$$

This is the current drawn by each core from the supply with no load connected.

Final Answer: $C_s = 4.2 \ \mu$F, $C_c = 1.6 \ \mu$F, $C_N = 9.0 \ \mu$F per phase; charging current $I_c = 107.7$ A.

3. Keyword Explanation

1. Belted cable

A three-core cable whose cores are insulated separately and then wrapped together in a common belt of insulation inside one sheath; used up to about 22 kV.

2. Charging current

The current that charges and discharges the cable's capacitance every cycle; it flows even with no load.

4. Notes and Remarks

- **Check with another test:** the capacitance between two cores, with the third left free, is $C_c + (C_c + C_s)/2 = 4.5 \ \mu$F, which equals $C_N/2$, as it should (two capacitances C_N in series).
- **Why the phase voltage?** C_N is a capacitance from a core to the neutral, so it carries the phase voltage $V_L/\sqrt{3}$.
- **Size of the current:** 107.7 A is large; the 9 µF values belong to a long cable, which is why long AC cables need shunt reactors, and long cable links use HVDC.

5. Final Concise Answer

(a) Skin effect is the crowding of an alternating current towards the surface of a conductor, because the inner filaments are linked by more flux; it raises the AC resistance above the DC resistance and grows with frequency and conductor size.

(b) FACTS are power-electronic controllers that change V, X or δ in $P = V_1 V_2 \sin\delta / X$: shunt (SVC, STATCOM: voltage control, reactive support), series (TCSC, SSSC: lower X, more power, flow control, damping), combined (UPFC: control of V, X and δ together; IPFC between lines). Applications: more power over existing lines, flow control, voltage control, stability and oscillation damping.

(c) By Gauss's law $D_x = Q/(2\pi x)$, so $g_x = Q/(2\pi\varepsilon_0\varepsilon_r x)$; integrating from $d/2$ to $D/2$ gives $V = \dfrac{Q}{2\pi\varepsilon_0\varepsilon_r}\ln(D/d)$, hence $g_x = V/[x\ln(D/d)]$, $g_{max} = 2V/[d\ln(D/d)]$ at the core and $g_{min} = 2V/[D\ln(D/d)]$ at the sheath.

(d) $C_s = 12.6/3 = 4.2 \ \mu$F; $C_c = (7.4 - 4.2)/2 = 1.6 \ \mu$F; $C_N = C_s + 3C_c = 9.0 \ \mu$F; $I_c = 2\pi \times 50 \times 9 \times 10^{-6} \times 66\,000/\sqrt{3} = 107.7$ A.

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4. (a) Define critical disruptive voltage. Write down its significance. Marks: 5
   (b) What is String Efficiency? Discuss the methods of improving String Efficiency. Marks: 10
   (c) The maximum and minimum stresses in the dielectric of a single core cable are 40 kV/cm (r.m.s.) and 10 kV/cm (r.m.s.) respectively. If the conductor diameter is 2 cm, find : (i) thickness of insulation (ii) operating voltage Marks: 10

4.(a) Define critical disruptive voltage. Write down its significance. Marks: 5

1. Theory

The basic picture (bare conductors in air, the field strongest at the conductor surface, air ionising above g_0, and the resulting corona) is built in 2(a). Here we find the voltage at which corona starts.

1. The field at the surface (Core Idea)

For a line with conductors of radius r and spacing d, with voltage V from each conductor to neutral, the field at distance x from a conductor's centre is $g = \dfrac{V}{x \ln(d/r)}$ (2(a)). At the surface, $x = r$:

$$g_{max} = \frac{V}{r \ln(d/r)}$$

2. Setting the surface field equal to the strength of air

Corona starts when g_{max} reaches the dielectric strength of air, g_0. Putting $g_{max} = g_0$ and solving for V:

$$V_c = g_0 \, r \, \ln \frac{d}{r}$$

3. Two corrections for real air and real conductors

- **Air density factor** $\delta = \dfrac{3.92\,b}{273 + t}$, with b the pressure in cm of mercury and t the temperature in °C. At 76 cm and 25 °C, $\delta = 1$. Thin air (high places, hot days) breaks down more easily, so $\delta < 1$ lowers V_c.
- **Irregularity factor** m_0: 1 for a smooth polished conductor, 0.98 to 0.92 for a roughened one, 0.87 to 0.80 for a stranded conductor. Dirt and sharp edges raise the local field and make corona start earlier.

Unit check: g_0 in kV/cm times r in cm gives kV; $\ln(d/r)$ has no unit.

2. Answer

Definition: The **critical disruptive voltage** V_c is the minimum phase-to-neutral voltage at which the electric field at the surface of the conductor becomes equal to the dielectric strength of the surrounding air (g_0 = 30 kV/cm peak = 21.2 kV/cm r.m.s. at 76 cm Hg and 25 °C), so that the air next to the conductor begins to ionise and **corona starts**.

Since the field at the conductor surface is $g_{max} = \dfrac{V}{r \ln(d/r)}$, putting $g_{max} = g_0$ gives $V_c = g_0 r \ln(d/r)$. Allowing for the air density and the surface condition (V.K. Mehta):

$$\boxed{V_c = m_0 \, g_0 \, \delta \, r \, \ln \frac{d}{r} \ \text{ kV r.m.s. (phase to neutral)}}$$

where

- $g_0 = 21.2$ kV/cm r.m.s., the dielectric strength of air
- $\delta = \dfrac{3.92\,b}{273 + t}$, the air density factor (b in cm Hg, t in °C)
- m_0, the irregularity factor (1 polished, 0.98–0.92 rough, 0.87–0.80 stranded)

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- r, the conductor radius (cm), and d the spacing between conductors (cm)

Significance:

1. **Onset of corona.** Below V_c the air round the conductor stays an insulator; above it, corona appears, with **power loss** (rising rapidly, as $(V - V_c)^2$ in Peek's formula), **radio and TV interference**, a hissing noise and **ozone**, which corrodes the conductor.
2. **Design of the line.** The working phase voltage of a line is kept **below** V_c in fair weather. So V_c fixes the **minimum conductor diameter** and the **spacing** for a given voltage. This is why EHV lines use large-diameter, hollow or **bundled** conductors (1(b)), which raise the effective radius.
3. **Weather and altitude.** V_c falls with δ and m_0: a line that is corona-free at sea level may show corona on a hill, on a hot day or in rain, fog and snow. The design margin must allow for this.
4. **Visual corona** starts at a slightly higher voltage, $V_v = m_v g_0 \delta r \left(1 + \dfrac{0.3}{\sqrt{\delta r}}\right) \ln \dfrac{d}{r}$; between V_c and V_v the corona is present but cannot be seen.

3. Keyword Explanation

1. Air density factor (δ)

The ratio of the air density to its value at 76 cm Hg and 25 °C; lower density means easier breakdown.

2. Irregularity factor (m_0)

A factor below 1 that accounts for the surface roughness and stranding of the conductor.

4. Notes and Remarks

- **V.K. Mehta** rounds g_0 to 21.1 kV/cm r.m.s.; $30/\sqrt{2} = 21.21$ is used here.
- **Line voltage:** V_c is a phase-to-neutral value; the line voltage at which corona starts in a three-phase line is $\sqrt{3}\,V_c$.
- The same definition is asked in the 2021-22 paper (1(a)), as extra reading.

4.(b) What is String Efficiency? Discuss the methods of improving String Efficiency. Marks: 10

1. Theory

1. Suspension insulators and strings (Core Idea)

Above about 33 kV a single pin insulator would have to be very large and heavy. Instead the conductor **hangs** below the cross-arm from a chain of identical porcelain or glass **discs** linked by metal caps and pins. The chain is a **string** and each disc is a **unit**. A higher voltage simply needs more discs.

2. Two kinds of capacitance

Each disc is a small capacitor: its metal cap and pin are two conductors with porcelain between them. This is the **self (mutual) capacitance** C of the disc. Each metal link between two discs is also close to the earthed tower, so there is a small capacitance from each link to the tower: the **shunt capacitance** C_1. Their ratio is

$$K = \frac{C_1}{C} = \frac{\text{shunt capacitance}}{\text{self capacitance}}$$

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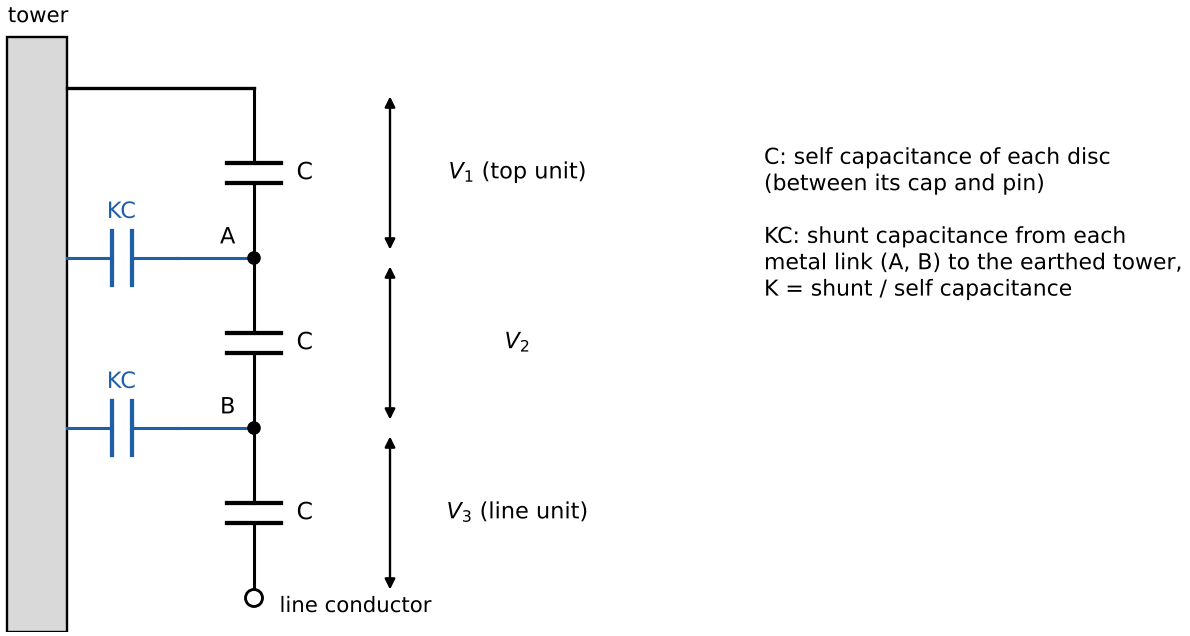


Figure 19: Equivalent circuit of a 3-disc string: self capacitance C for each disc, shunt capacitance KC from each link to the tower (reused from the 2021-22 paper)

Reading the figure. The tower (earth) is on the left. The three discs are the capacitors C in series, from the tower (top unit) down to the line conductor (line unit). From each metal link, A and B, a shunt capacitor KC leads to the tower.

3. Why the voltage is not shared equally

If there were no shunt capacitance ($K = 0$), the same current would pass through every disc, and each would take an equal share of the voltage. But at every link part of the current leaks to the tower through KC. So the disc **nearest the line** carries its own current plus all the leakage currents of the links above it: it has the most current and therefore the **largest voltage**. The top disc, nearest the tower, has the least. The string can only be used up to the voltage at which the **line unit** is just safe; the other discs then work below their rating.

2. Answer

String efficiency is the ratio of the voltage across the whole string to n times the voltage across the unit nearest the line conductor:

$$\eta_{string} = \frac{\text{voltage across the string}}{n \times \text{voltage across the unit nearest the conductor}} = \frac{V}{n\,V_n} \times 100\ \%$$

where n is the number of units. It is 100 % when all units share the voltage equally; the more unequal the sharing, the lower the efficiency.

Why it is less than 100 % (three-unit string, figure above). Let V_1, V_2, V_3 be the voltages across the top, middle and line units, ω the angular frequency, and V the phase voltage across the string. A capacitor C with voltage V carries current ωCV.

At link A (KCL: current in unit 2 = current in unit 1 + shunt current at A):

$$\omega C V_2 = \omega C V_1 + \omega K C V_1 \quad \Rightarrow \quad V_2 = (1+K)V_1$$

At link B (B is at $V_1 + V_2$ from the tower):

$$\omega C V_3 = \omega C V_2 + \omega K C (V_1 + V_2) \quad \Rightarrow \quad V_3 = (1 + 3K + K^2)V_1$$

$$V = V_1 + V_2 + V_3 = (3 + 4K + K^2)V_1, \qquad \eta = \frac{V}{3V_3} = \frac{3 + 4K + K^2}{3(1 + 3K + K^2)}$$

For example, $K = 0.1$ gives shares of 29.3 %, 32.3 % and 38.4 % of V, and $\eta = 86.8$ %; with four units

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η falls to 76.6 %. The efficiency is improved by making the shares more equal, in three ways:

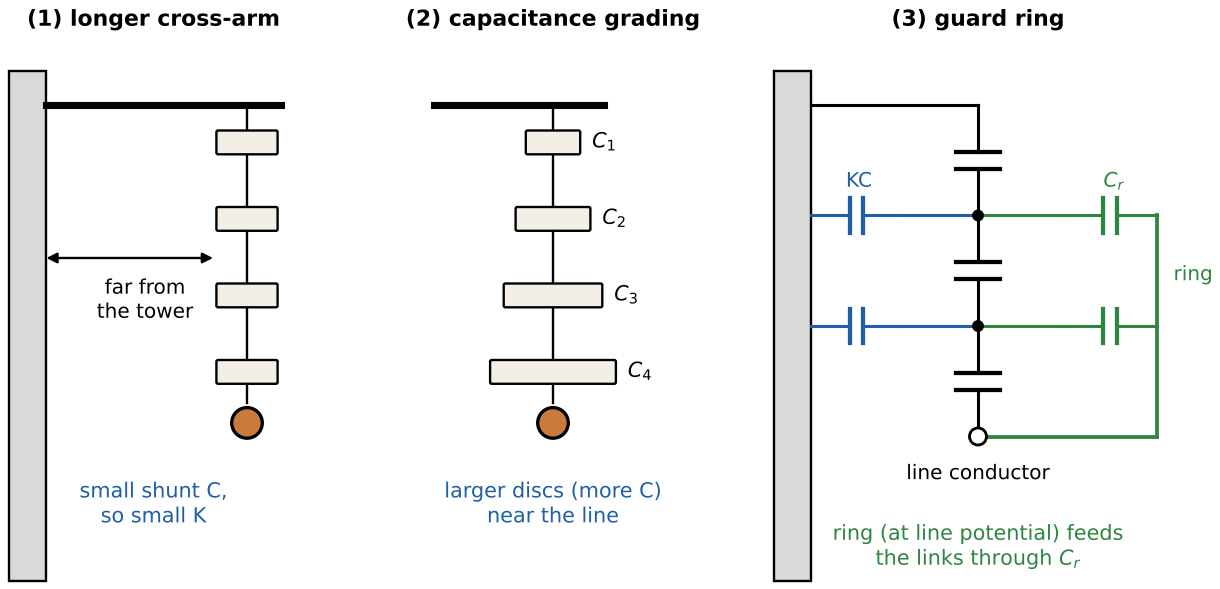


Figure 20: Three ways of improving string efficiency: (1) a longer cross-arm (smaller K), (2) capacitance grading (larger discs near the line), (3) a guard ring connected to the line

1. Longer cross-arms (reducing K). The shunt capacitance falls when the string hangs further from the tower, so K becomes smaller and the shares become more equal. The cost and the strength of the tower limit the length of the cross-arm, so in practice K cannot be reduced much below about 0.1.

2. Grading the insulators (capacitance grading). Discs of **different capacitance** are used: the top unit (near the tower) has the smallest capacitance and the line unit the largest. A disc with a larger capacitance carries its larger current at a smaller voltage, so the voltages can be made equal. For equal voltages v across a three-unit string:

$$\text{at A: } C_2 v = C_1 v + KC_1 v \Rightarrow C_2 = C_1(1+K); \qquad \text{at B: } C_3 v = C_2 v + KC_1(2v) \Rightarrow C_3 = C_1(1+3K)$$

The drawback is that many different disc sizes must be made and stocked.

3. Guard ring (static shielding). A metal ring, connected to the **line conductor**, surrounds the bottom unit. It forms a capacitance C_r from the ring (at line potential) to each link. The charging current that flows through C_r into each link supplies the current that leaks from the link to the tower through KC. If the two are equal at every link, the same current flows through all units and the voltage is shared equally (η close to 100 %). The guard ring also acts as an **arcing ring**, keeping flash-over arcs away from the porcelain.

Conclusion: the string efficiency is high when K is small and the string is short; it is improved by longer cross-arms, by capacitance grading and by a guard ring (the most used in practice, together with arcing horns).

3. Keyword Explanation

1. Self capacitance (C)

The capacitance of one disc between its metal cap and pin.

2. Shunt capacitance (KC)

The capacitance between a metal link of the string and the earthed tower.

3. Guard ring

A metal ring around the line-end unit, connected to the conductor, that equalises the voltage sharing by static shielding.

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4. Notes and Remarks

- **General rule** for n units: $V_{k+1} = V_k + K(V_1 + V_2 + \cdots + V_k)$; η falls as K or n grows. For $K = 0.1$: 86.8 % (3 units), 76.6 % (4), 66.9 % (5).
- **Wet weather** makes the sharing worse, because leakage over the wet surfaces adds to the shunt currents.
- **Check:** $K = 0$ gives $\eta = 3/3 = 100$ %.
- The derivation, with the effect of K and of the number of discs, is also in 3(d) of the 2021-22 paper, as extra reading.

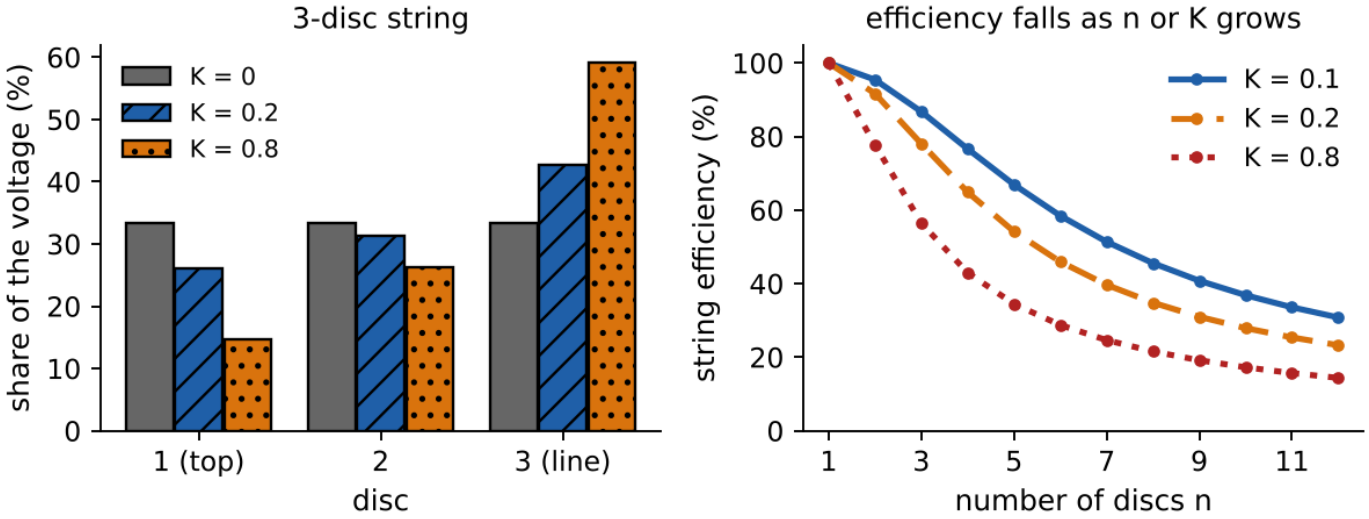


Figure 21: String efficiency against the number of discs for three values of K (reused from the 2021-22 paper); left, the share of each disc of a 3-disc string

Reading the figure. On the right, each curve falls as discs are added: solid blue is $K = 0.1$, dashed orange $K = 0.2$ and dotted red $K = 0.8$. A larger K gives a lower efficiency for every string length.

4.(c) The maximum and minimum stresses in the dielectric of a single core cable are 40 kV/cm (r.m.s.) and 10 kV/cm (r.m.s.) respectively. If the conductor diameter is 2 cm, find : (i) thickness of insulation (ii) operating voltage Marks: 10

1. Theory

From 3(c), the stress in the insulation of a single-core cable at distance x from the centre is

$$g = \frac{V}{x \ln(D/d)}$$

with its largest value at the core ($x = d/2$) and its smallest at the sheath ($x = D/2$):

$$g_{max} = \frac{2V}{d \ln(D/d)}, \qquad g_{min} = \frac{2V}{D \ln(D/d)}$$

1. The ratio gives the sheath diameter (Core Idea)

Dividing the two, everything cancels except the diameters:

$$\frac{g_{max}}{g_{min}} = \frac{D}{d}$$

So if both stresses and the core diameter are known, the sheath diameter follows at once: $D = d\, g_{max}/g_{min}$.

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2. Insulation thickness

The insulation fills the ring between the core (radius $d/2$) and the sheath (radius $D/2$), so its thickness is

$$t = \frac{D - d}{2}$$

3. The voltage

Once D is known, either stress formula gives the voltage: $V = g_{max}\frac{d}{2}\ln\frac{D}{d}$.

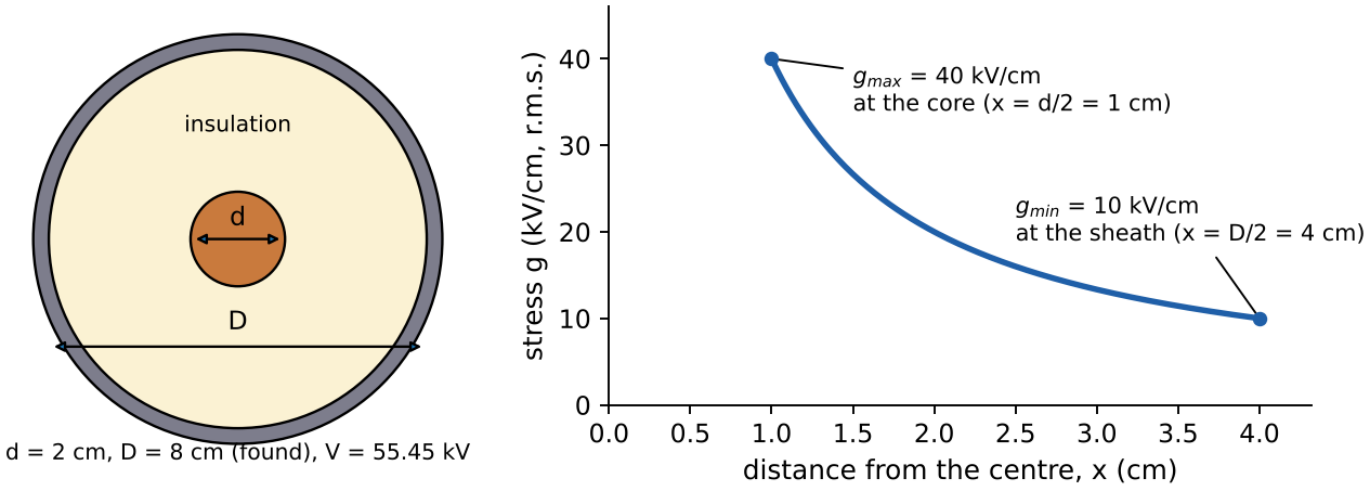


Figure 22: The cable of this question: d = 2 cm, D = 8 cm; the stress falls from 40 kV/cm at the core to 10 kV/cm at the sheath

Reading the figure. The solid blue curve falls as $1/x$ from 40 kV/cm at $x = 1$ cm (the core surface) to 10 kV/cm at $x = 4$ cm (the sheath). The ratio $40/10 = 4$ is the ratio of the radii, which fixes $D = 8$ cm.

2. Answer

Given:

- $g_{max} = 40$ kV/cm r.m.s., $g_{min} = 10$ kV/cm r.m.s.
- Conductor diameter $d = 2$ cm

Relationships:

$$\frac{g_{max}}{g_{min}} = \frac{D}{d}, \qquad t = \frac{D - d}{2}, \qquad V = g_{max}\frac{d}{2}\ln\frac{D}{d}$$

Step 1: Sheath diameter

$$D = d\,\frac{g_{max}}{g_{min}} = 2 \times \frac{40}{10} = 8 \text{ cm}$$

Step 2: (i) Thickness of insulation

$$t = \frac{D - d}{2} = \frac{8 - 2}{2} = 3 \text{ cm}$$

$$\boxed{t = 3 \text{ cm}}$$

Step 3: (ii) Operating voltage

$$V = g_{max}\frac{d}{2}\ln\frac{D}{d} = 40 \times \frac{2}{2} \times \ln 4 = 40 \times 1.3863 = 55.45 \text{ kV}$$

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$$\boxed{V = 55.45 \text{ kV r.m.s.}}$$

This is the voltage between the core and the earthed sheath. Check with g_{min}: $V = g_{min}\frac{D}{2}\ln\frac{D}{d} = 10 \times 4 \times 1.3863 = 55.45$ kV, the same.

Final Answer: (i) insulation thickness 3 cm (sheath inside diameter 8 cm); (ii) operating voltage 55.45 kV r.m.s. between core and sheath.

3. Keyword Explanation

1. Operating voltage

The working voltage of the cable, here between the core and the sheath, at which the stress at the core is the given g_{max}.

4. Notes and Remarks

- **Peak voltage:** $55.45 \times \sqrt{2} = 78.42$ kV.
- **In a three-phase system** this core-to-sheath (phase) voltage corresponds to a line voltage of $55.45 \times \sqrt{3} = 96.0$ kV.

5. Final Concise Answer

(a) V_c is the minimum phase-to-neutral voltage at which the field at the conductor surface reaches the strength of air (30 kV/cm peak), so corona starts: $V_c = m_0 g_0 \delta r \ln(d/r)$ kV. Significance: corona loss, interference and ozone appear above it; the line is designed to work below it, which fixes the minimum conductor size and spacing (hence large and bundled conductors); it falls in thin air, hot or wet weather.

(b) $\eta = V/(nV_n)$; shunt capacitance KC makes the line unit take the most voltage ($V_2 = (1+K)V_1$, $V_3 = (1+3K+K^2)V_1$; $K = 0.1$ gives 86.8 % for 3 units). Improved by longer cross-arms (smaller K), capacitance grading ($C_2 = C_1(1+K)$, $C_3 = C_1(1+3K)$) and a guard ring at line potential that supplies the shunt currents.

(c) $D = d\, g_{max}/g_{min} = 8$ cm; (i) thickness $(D-d)/2 = 3$ cm; (ii) $V = g_{max}(d/2)\ln(D/d) = 40 \times 1 \times \ln 4 = 55.45$ kV r.m.s.

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5. (a) What are the causes of insulation failure? Marks: 5

(b) Varley loop test is performed to locate an earth fault on a 20 km long cable. The resistance per km of the single conductor is 20 ohm. The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 ohm. The fixed resistors have equal values. Calculate the distance of the fault from the test end. Marks: 10

(c) A 132 KV transmission line has the following data: Weight of the conductor= 680 kg/km, Length of span= 260 m, Ultimate strength= 3100 kg, Safety Factor= 2. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 meters. Marks: 10

5.(a) What are the causes of insulation failure? Marks: 5

1. Theory

1. What an insulator has to do (Core Idea)

The conductors of an overhead line are fixed to the earthed towers or poles through **insulators**, made of porcelain, glass or a polymer. An insulator has two jobs at once:

- **electrical:** stop current from leaking from the conductor to the earthed support, even when wet and dirty, and withstand lightning and switching surges;
- **mechanical:** carry the weight and the tension of the conductor, and the loads of wind and ice.

It fails when it cannot do one of these jobs any longer.

2. Two ways the insulation can break down

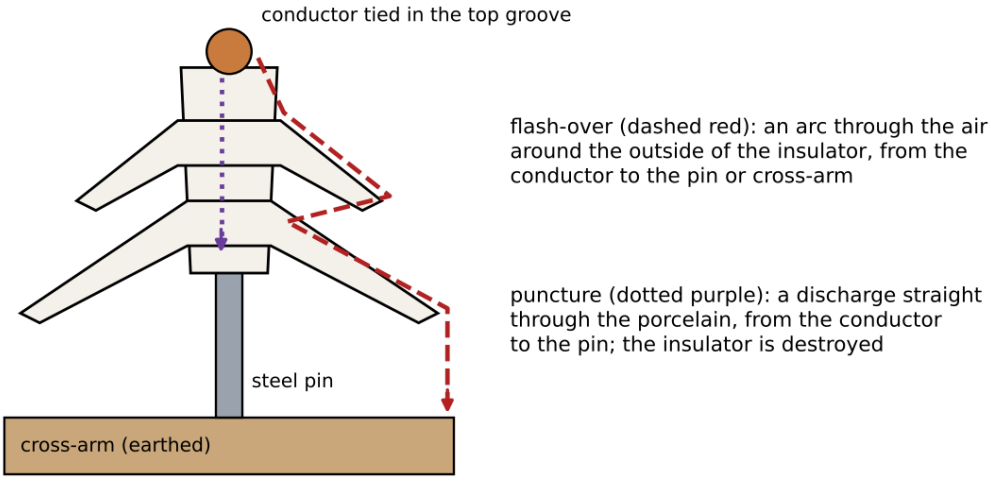


Figure 23: A pin insulator: flash-over (dashed red) is an arc through the air around the insulator; puncture (dotted purple) is a discharge through the porcelain itself (reused from the 2021-22 paper)

Reading the figure. The conductor is tied in the top groove; the steel pin goes down to the earthed cross-arm. The dashed red path is a **flash-over**: the air round the insulator breaks down and an arc jumps over the sheds to the pin. The dotted purple path is a **puncture**: a discharge goes straight through the porcelain. A puncture destroys the insulator; a flash-over usually does not, but its heat can crack it.

3. Why porcelain can fail

Porcelain is strong in compression but brittle; it is joined to metal caps and pins by cement, and these three materials expand by different amounts when they are heated. Its body must be dense (not porous) and its surface glazed, so that water and dirt do not stay on it or soak into it.

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2. Answer

Assumption: "insulation failure" is read as the failure of the **insulators of overhead lines** (V.K. Mehta, "Causes of insulator failure"), the topic of the questions beside it; the breakdown of cable insulation is summarised in the Notes.
The main causes of failure of overhead-line insulators (V.K. Mehta) are:

1. **Cracking of the insulator.** Porcelain, the steel fittings and the cement expand and contract by different amounts with temperature. The repeated mechanical stress cracks the porcelain, and moisture entering the crack lets current leak or puncture.
2. **Defective insulating material.** Impurities, air pockets or poor firing in the porcelain make weak spots, where the insulator punctures at a voltage far below its rating.
3. **Porosity of the material.** Porous porcelain absorbs moisture; its insulation resistance falls, leakage current flows, and the insulator heats and fails.
4. **Improper glazing.** Where the glaze is missing or damaged, moisture and dust settle on the surface and form leakage paths over it, leading to flash-over.
5. **Flash-over.** An arc from the conductor to the pin or cross-arm, caused by surges or by a dirty wet surface. Its heat can shatter or crack the insulator.
6. **Mechanical stresses.** Excessive conductor tension, wind and ice loads, or weak spots in manufacture break the insulator mechanically.
7. **Short circuits** across the insulator by birds, animals or tree branches.

Besides these, **ageing** (slow loss of strength with time, weather and pollution) and **puncture** by a surge that is too high for the porcelain end the life of an insulator. Good material, proper glazing, a **safety factor** (puncture strength / flash-over voltage, about 10) and regular cleaning and testing reduce failures.

3. Keyword Explanation

1. Flash-over

Breakdown of the air around the insulator surface, so that an arc joins the conductor to the earthed fitting.

2. Puncture

Breakdown through the body of the insulator; it destroys the insulator.

4. Notes and Remarks

- **If "insulation" means cable insulation:** the causes are moisture entering through a damaged sheath, voids (air pockets) in the dielectric that ionise and erode it, **thermal breakdown** (dielectric loss heating the insulation, which then conducts more), overstress at the core from overvoltage, ageing, and mechanical damage during laying.
- **Testing:** insulators are given flash-over, puncture and mechanical tests before use; the puncture voltage must be well above the flash-over voltage, so that a flash-over (which the insulator survives) happens first.

5.(b) Varley loop test is performed to locate an earth fault on a 20 km long cable. The resistance per km of the single conductor is 20 ohm. The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 ohm. The fixed resistors have equal values. Calculate the distance of the fault from the test end. Marks: 10

1. Theory

1. Why a loop test? (Core Idea)

An underground cable is buried, so a damaged spot cannot be seen. If the insulation breaks at one point, current leaks from the core to the earth there: a **fault to earth**. To repair it with little digging, we must know its distance from the end. **Loop tests** find it by measuring resistances with a bridge.

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2. The Wheatstone bridge

Four resistors form a square, with a battery across one diagonal and a sensitive current meter (a **galvanometer** G) across the other. The resistors are adjusted until G reads zero. Then the two ends of G are at the same voltage, which happens only when the resistors are in proportion:

$$\frac{P}{Q} = \frac{R}{S}$$

where P, Q are the two arms on one side of G and R, S the two on the other.

3. The Varley loop

The faulty conductor is joined at the far end to a **healthy** (sound) conductor laid beside it, so the two form one long **loop**. The ratio arms P and Q are **fixed**; balance is obtained with a **variable resistance** S in series with the faulty conductor at the test end.

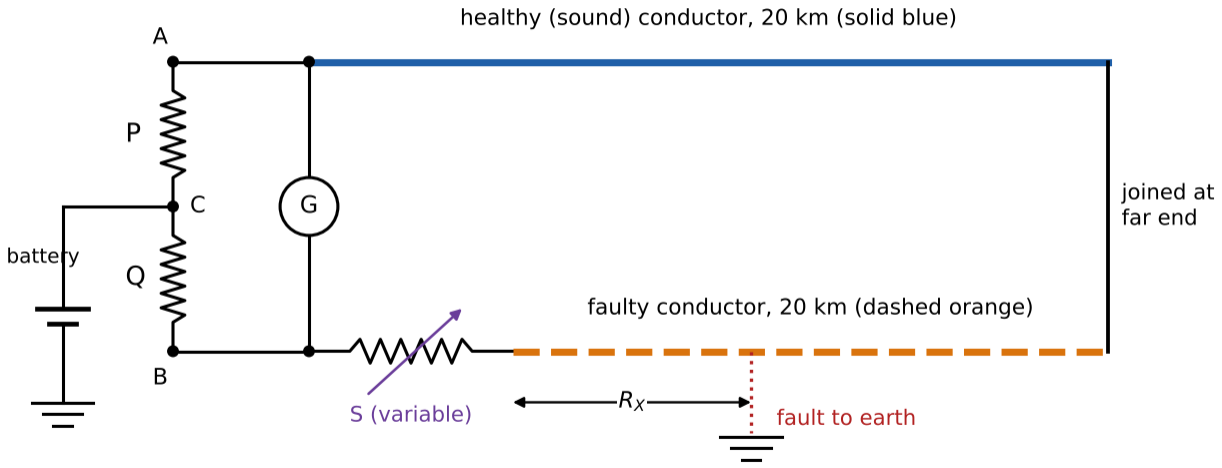


Figure 24: Varley loop: fixed ratio arms P and Q, galvanometer G between A and B, battery from C to earth, the variable resistance S in series with the faulty conductor, and the two conductors joined at the far end

Reading the figure. The battery current goes from C through P and Q, along the conductors to the fault, and back through the earth. So the fault (earth) is the fourth corner of the bridge. The four arms are:

- C–A: P;
- C–B: Q;
- A–fault: the whole healthy conductor plus the far part of the faulty one, $R_{loop} - R_X$;
- B–fault: the variable resistance and the faulty conductor up to the fault, $S + R_X$.

Here R_{loop} is the resistance of the whole loop and R_X that of the faulty conductor from the test end to the fault.

4. The balance condition

When G reads zero:

$$\frac{P}{Q} = \frac{R_{loop} - R_X}{R_X + S} \quad \Rightarrow \quad R_X = \frac{Q\,R_{loop} - P\,S}{P + Q}$$

With equal ratio arms ($P = Q$) this becomes simply

$$R_X = \frac{R_{loop} - S}{2}$$

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The fault resistance sits in the battery branch, so it does not upset the balance. If R_{loop} is not known, a first balance with the battery connected across the loop instead of to earth measures it; here it follows from the data.

2. Answer

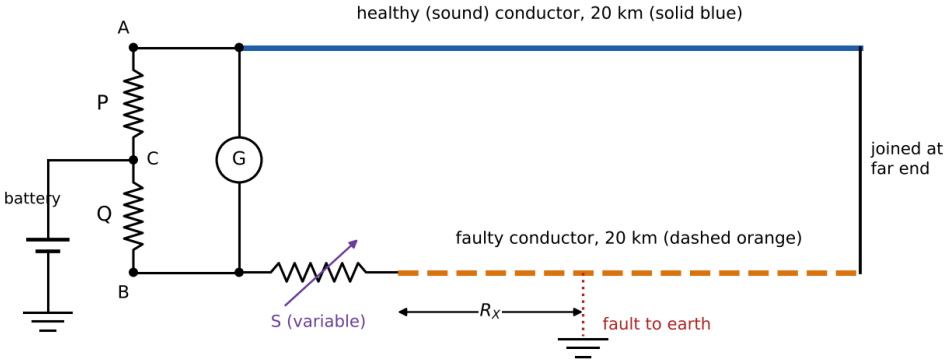


Figure 25: Repeated from the Theory: the Varley loop with S in series with the faulty conductor

Given:

- Cable length $L = 20$ km; resistance of one conductor $r = 20\ \Omega$/km
- Loop completed with a similar healthy conductor
- Variable resistance in series with the faulty conductor at balance: $S = 200\ \Omega$
- Fixed ratio arms equal: $P = Q$

Relationship (at balance):

$$\frac{P}{Q} = \frac{R_{loop} - R_X}{R_X + S} \quad \Rightarrow \quad R_X = \frac{R_{loop} - S}{2} \quad (P = Q)$$

Step 1: Resistance of the loop

$$R_{loop} = 2 \times L \times r = 2 \times 20 \times 20 = 800\ \Omega$$

Step 2: Resistance up to the fault

$$R_X = \frac{800 - 200}{2} = 300\ \Omega$$

$$\boxed{R_X = 300\ \Omega}$$

This is the resistance of the faulty conductor from the test end to the fault.

Step 3: Distance of the fault

$$X = \frac{R_X}{r} = \frac{300}{20} = 15\ \text{km}$$

$$\boxed{X = 15\ \text{km from the test end}}$$

The fault is 15 km along the 20 km cable, a possible place, so the answer is sensible.
Final Answer: the fault is **15 km** from the test end.

3. Keyword Explanation

1. Loop test

A bridge measurement in which the faulty conductor and a sound conductor, joined at the far end, form two arms of a Wheatstone bridge.

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2. Galvanometer

A very sensitive current meter used to detect the balance (zero current) of a bridge.

4. Notes and Remarks

- **If S were in series with the healthy conductor** instead, the balance would give $R_X = (R_{loop} + S)/2 = 500\ \Omega$, i.e. 25 km: longer than the cable, so impossible. The paper's "connected to the faulty conductor" fits the result of 15 km.
- **Murray loop:** the related test balances by varying the ratio arms and has no S: $R_X = \dfrac{Q}{P+Q} R_{loop}$ (solved in 1(b) of the 2021-22 paper, as extra reading). The Varley loop also measures R_{loop} itself, which is useful when the conductor resistance is not known exactly.

5.(c) A 132 KV transmission line has the following data: Weight of the conductor= 680 kg/km, Length of span= 260 m, Ultimate strength= 3100 kg, Safety Factor= 2. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 meters. Marks: 10

1. Theory

1. Why a line sags (Core Idea)

A conductor strung between two supports cannot be pulled perfectly straight: that would need an infinite tension. It hangs in a curve, lowest at the middle. The vertical distance between the supports' level and the lowest point is the **sag**. A small sag means a large tension (the conductor may break, especially when cold weather shortens it); a large sag means taller supports to keep the conductor clear of the ground. The sag is chosen to balance the two.

2. Sag with supports at the same level

Take the lowest point O as the origin. A conductor of weight w per metre is held at O by the horizontal tension T. For the usual small sags the curve is a parabola, and taking moments about a point at distance x from O gives its height $y = \dfrac{wx^2}{2T}$. At a support, $x = l/2$, so the sag is

$$S = \frac{w\,l^2}{8\,T}$$

where l is the span (m), w the weight per metre and T the tension (both in kg-wt, or both in N: only their ratio matters).

3. Working tension and safety factor

The conductor must never be stressed near its breaking point. The **working tension** is the ultimate (breaking) strength divided by a **safety factor** (usually 2):

$$T = \frac{\text{ultimate strength}}{\text{safety factor}}$$

4. Height of the supports

The lowest point of the conductor must stay at least the required **ground clearance** above the ground, so the conductor must be attached to the supports at

$$h = \text{clearance} + S$$

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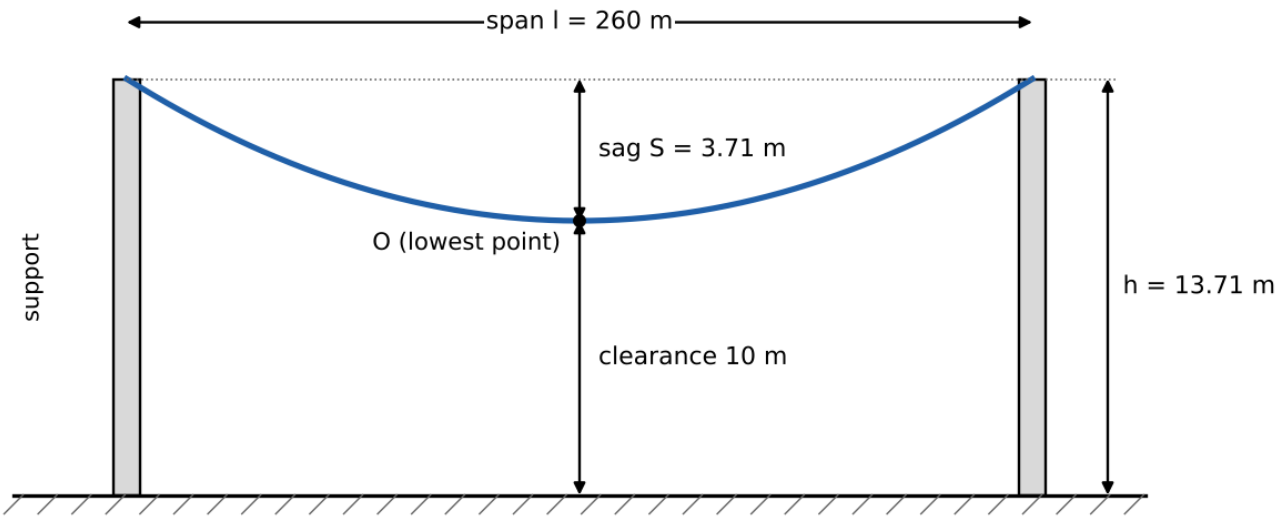


Figure 26: Conductor between two supports at the same level: sag S below the support level, ground clearance below the lowest point O, and the height h of the supports

Reading the figure. The solid blue curve is the conductor between two supports 260 m apart. Its lowest point O is $S = 3.71$ m below the support level, and must stay 10 m above the ground; so the conductor is fixed at $h = 13.71$ m. (The sag is drawn larger than to scale.)

2. Answer

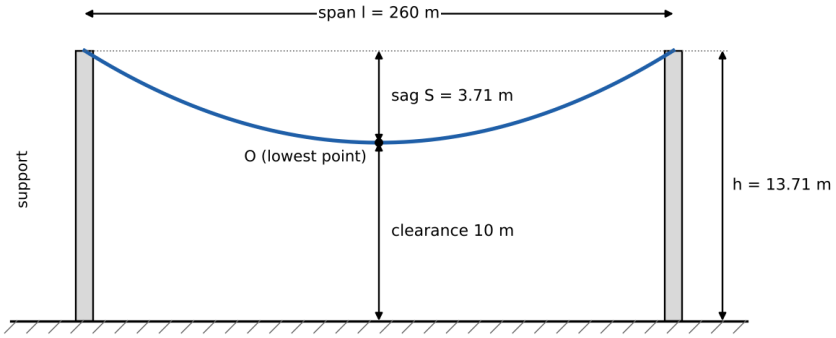


Figure 27: Repeated from the Theory: supports at the same level, sag S and clearance

Given:

- Weight of conductor $w = 680$ kg/km $= 0.68$ kg/m
- Span $l = 260$ m
- Ultimate strength 3100 kg; safety factor 2
- Ground clearance 10 m; supports at the same level (assumed, as nothing else is stated); the 132 kV rating is not needed for the sag

Relationships:

$$T = \frac{\text{ultimate strength}}{\text{safety factor}}, \qquad S = \frac{w\,l^2}{8\,T}, \qquad h = \text{clearance} + S$$

Step 1: Working tension

$$T = \frac{3100}{2} = 1550 \text{ kg}$$

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Step 2: Sag

$$S = \frac{0.68 \times 260^2}{8 \times 1550} = \frac{0.68 \times 67\,600}{12\,400} = \frac{45\,968}{12\,400} = 3.707 \text{ m}$$

$$\boxed{S = 3.71 \text{ m}}$$

Step 3: Height of the supports

$$h = 10 + 3.707 = 13.707 \text{ m}$$

$$\boxed{h = 13.71 \text{ m above the ground}}$$

Final Answer: sag 3.71 m; the conductor should be supported at **13.71 m** above the ground.

3. Keyword Explanation

1. Sag

The vertical distance between the level of the supports and the lowest point of the conductor.

2. Safety factor

The ratio of the ultimate (breaking) strength to the working tension; 2 means the conductor works at half its breaking load.

4. Notes and Remarks

- **Units:** kg-wt is used for both w and T (as in V.K. Mehta), so $g = 9.81$ cancels. In newtons: $w = 6.671$ N/m, $T = 15\,206$ N, and the sag is the same 3.707 m.
- **Wind and ice** would add to the effective weight ($w_{eff} = \sqrt{(w + w_i)^2 + w_w^2}$) and increase the sag; none is given here.
- **Unequal support levels** (hilly ground) need a different formula, with the lowest point no longer at the middle of the span.

5. Final Concise Answer

(a) Causes of insulator failure: cracking from unequal expansion of porcelain, steel and cement; defective (impure) material; porosity (moisture absorbed); improper glazing (dirt and moisture on the surface); flash-over (arc heat cracks it); mechanical stresses (tension, wind, ice); short circuits by birds or branches; also ageing and puncture by surges.
(b) $R_{loop} = 2 \times 20 \times 20 = 800\ \Omega$; with $P = Q$, $R_X = (R_{loop} - S)/2 = (800 - 200)/2 = 300\ \Omega$; fault at $300/20 = 15$ km from the test end.
(c) $T = 3100/2 = 1550$ kg; $S = wl^2/(8T) = 0.68 \times 260^2/(8 \times 1550) = 3.71$ m; height $h = 10 + 3.71 = 13.71$ m.

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6. (a) Define Power Angle. Briefly explain Equal Area Criterion. Marks: 5

(b) Explain what happens for sudden increase in mechanical input to generator using Pe-δ diagram. Marks: 10

(c) The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus (|V|=1.0 pu), with the generator terminal voltage (|Vt|=1.0 pu). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (a) System healthy (b) One line shorted (3 phase) in the middle. Marks: 10

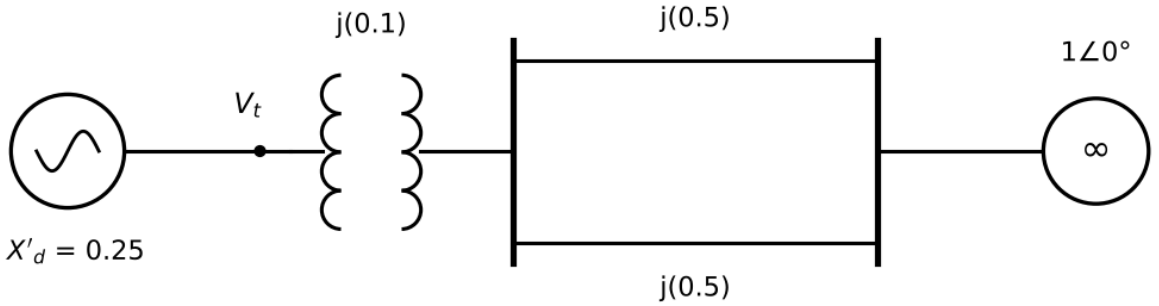


Figure 28: Fig. 2 of the paper (redrawn; the same figure as in the 2021-22 paper): figure for Q-6(c)

6.(a) Define Power Angle. Briefly explain Equal Area Criterion. Marks: 5

1. Theory

1. A generator connected to a large system (Core Idea)

A synchronous generator is turned by a turbine. The turbine gives **mechanical power** P_m to the rotor; the generator sends **electrical power** P_e into the network. Think of the generator as an internal emf E behind a reactance X (its own reactance plus that of the transformer and line), connected to a very large system whose voltage V and frequency never change: an **infinite bus**.

2. The power-angle equation

The power sent from $E\angle\delta$ through a pure reactance X to $V\angle 0$ is

$$P_e = \frac{EV}{X}\sin\delta = P_{max}\sin\delta$$

The angle δ is the angle by which the generator's emf (and its rotor) is ahead of the infinite bus. Plotting P_e against δ gives the **power-angle curve**: a sine wave, largest at $\delta = 90°$, where $P_{max} = EV/X$.

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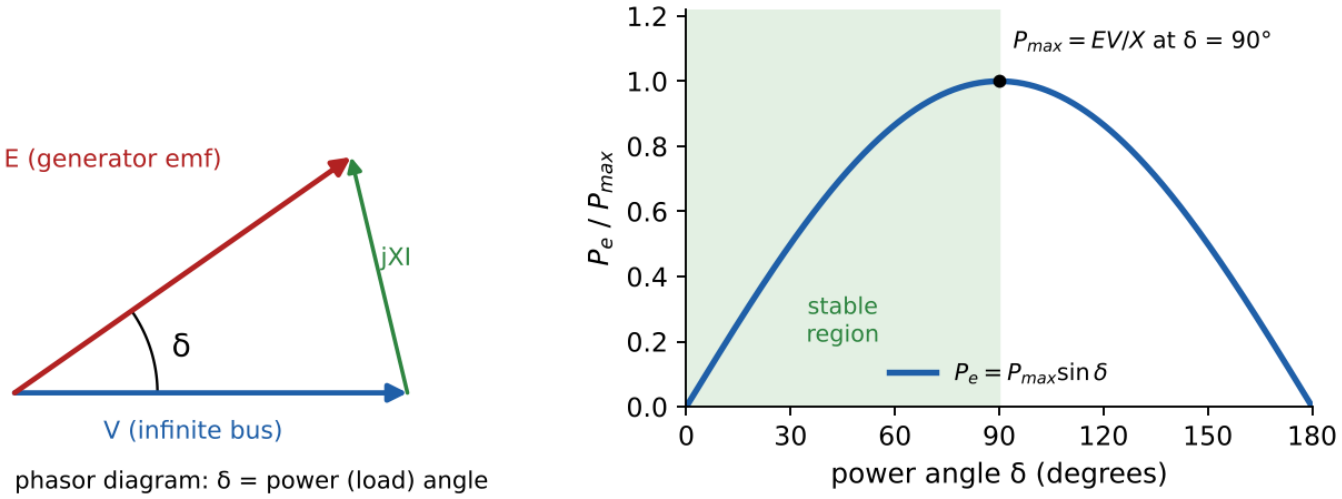


Figure 29: Left: phasor diagram, E leads V by the power angle delta, and E = V + jXI. Right: the power-angle curve, stable on the rising part (shaded)

Reading the figure. On the left, V (blue, the infinite bus) is the reference; the generator emf E (red) is ahead of it by δ; the green phasor jXI is the voltage across the reactance. On the right, the solid blue curve is $P_e/P_{max} = \sin\delta$. On the rising part ($0 < \delta < 90°$, shaded) a small increase of δ increases the power, which pulls the rotor back: these operating points are stable.

3. What speeds the rotor up or slows it down?

The rotor is a heavy spinning wheel. Newton's law for rotation gives the **swing equation**:

$$M\frac{d^2\delta}{dt^2} = P_m - P_e$$

where M is the inertia constant. If $P_m > P_e$ the rotor **accelerates** and δ grows; if $P_m < P_e$ it **decelerates**; if they are equal the speed and δ stay constant.

4. Energy gained and returned

Multiplying the swing equation by $2\,d\delta/dt$ and integrating gives

$$\left(\frac{d\delta}{dt}\right)^2 = \frac{2}{M}\int_{\delta_0}^{\delta}(P_m - P_e)\,d\delta$$

The rotor started at synchronous speed ($d\delta/dt = 0$). It stops swinging (back at synchronous speed) when the integral is zero again: the energy taken in while $P_m > P_e$ has all been given back while $P_e > P_m$. On the power-angle diagram the integral is an **area**.

2. Answer

Power angle: the **power angle (load or torque angle)** δ is the angle between the internal emf E of the synchronous generator and the voltage V of the infinite bus (equivalently, the angle by which the rotor leads the bus's reference axis). It fixes the power transferred, $P_e = \dfrac{EV}{X}\sin\delta$; the generator is stable for $0 < \delta < 90°$ in steady state, and $P_{max} = EV/X$ at $\delta = 90°$.

Equal area criterion: after a disturbance, a machine connected to an infinite bus stays in synchronism if the rotor angle stops increasing, that is, if

$$\int_{\delta_0}^{\delta_{max}}(P_m - P_e)\,d\delta = 0 \quad \Leftrightarrow \quad A_1 = A_2$$

- A_1, the **accelerating area**, is the area between the P_m line and the P_e curve where $P_m > P_e$: the kinetic energy the rotor gains.

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- A_2, the **decelerating area**, is the area where $P_e > P_m$: the energy it gives back.

The system is stable if the available decelerating area (up to the angle where P_e falls below P_m again) is **at least equal** to the accelerating area; otherwise δ keeps growing and synchronism is lost. The criterion gives the stability without solving the swing equation in time, and it gives the **critical clearing angle** of a fault.

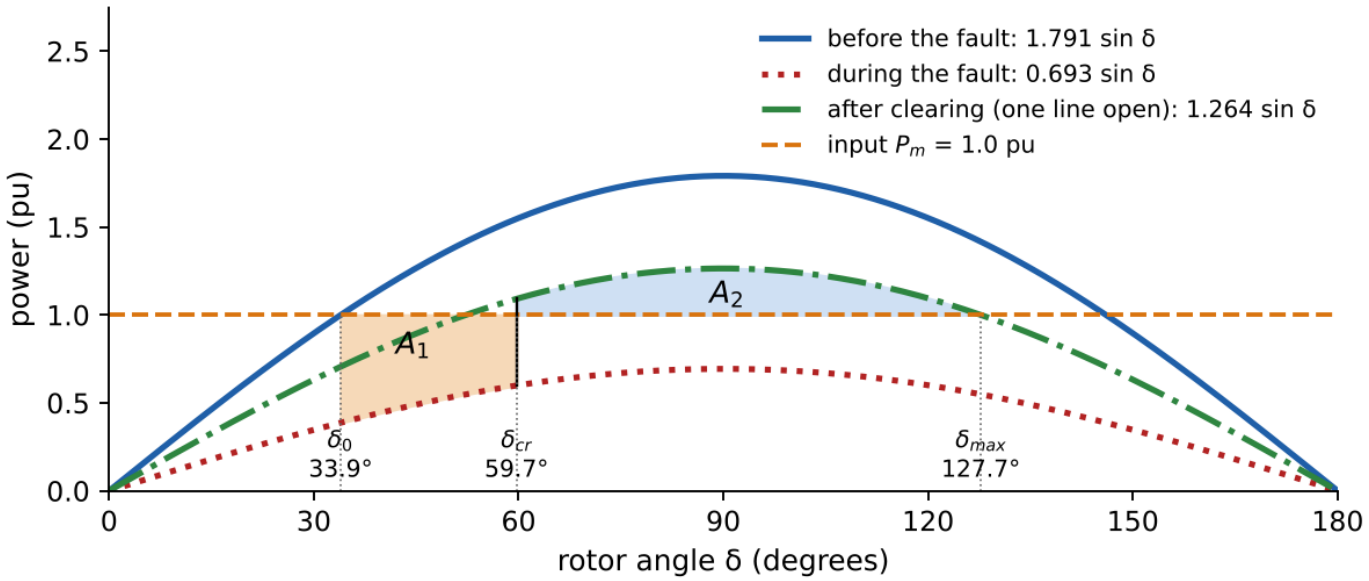


Figure 30: Equal area criterion for a fault cleared by opening the faulted line (numbers of 6(c)): before the fault (solid blue), during the fault (dotted red), after clearing (dash-dot green), input Pm (dashed orange); A1 = A2 at the critical clearing angle

Illustration. With the numbers of 6(c), the generator runs at $\delta_0 = 33.9°$. A fault drops the curve to $0.693 \sin\delta$, below the input of 1.0 pu, so the rotor accelerates (area A_1). When the faulted line is opened, the curve becomes $1.264 \sin\delta$, above the input, and the rotor decelerates (area A_2), up to at most $\delta_{max} = 127.7°$. Clearing at $\delta_{cr} = 59.7°$ makes $A_1 = A_2 = 0.224$ pu $\cdot$ rad exactly: this is the **critical clearing angle**. A later clearing makes A_1 larger than the area available, and the machine loses synchronism.

3. Keyword Explanation

1. Infinite bus

A system so large that its voltage and frequency do not change, whatever one generator does.

2. Swing equation

$M\, d^2\delta/dt^2 = P_m - P_e$: the rotor's equation of motion.

3. Critical clearing angle

The largest rotor angle at which a fault can be cleared with the machine still stable ($A_1 = A_2$).

4. Notes and Remarks

- **Where the criterion applies:** one machine against an infinite bus, or two machines; damping and the governor are neglected, so it is slightly pessimistic.
- **Critical clearing angle formula:** $\cos\delta_{cr} = \dfrac{P_m(\delta_{max} - \delta_0) + P_{3max}\cos\delta_{max} - P_{2max}\cos\delta_0}{P_{3max} - P_{2max}}$, with P_{2max} during and P_{3max} after the fault (angles in radians); with the numbers above it gives 59.7°.

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6.(b) Explain what happens for sudden increase in mechanical input to generator using Pe-δ diagram. Marks: 10

1. Theory

The power-angle curve $P_e = P_{max}\sin\delta$, the swing equation $M\,d^2\delta/dt^2 = P_m - P_e$ and the equal area criterion are built in 6(a). The key facts for this question:

1. The rotor cannot change its angle instantly (Core Idea)

The rotor is heavy, so δ cannot jump; it can only change through a change of speed. Right after a sudden change of P_m, the electrical output stays where it was, on the same point of the curve.

2. The rotor overshoots

When the rotor reaches the new balance point, it is running **faster** than synchronous speed (it has gained energy), so it does not stop there: it overshoots, like a pendulum passing its lowest point. It slows down only while $P_e > P_m$, and it returns to synchronous speed when the energy gained has been given back: the equal areas.

2. Answer

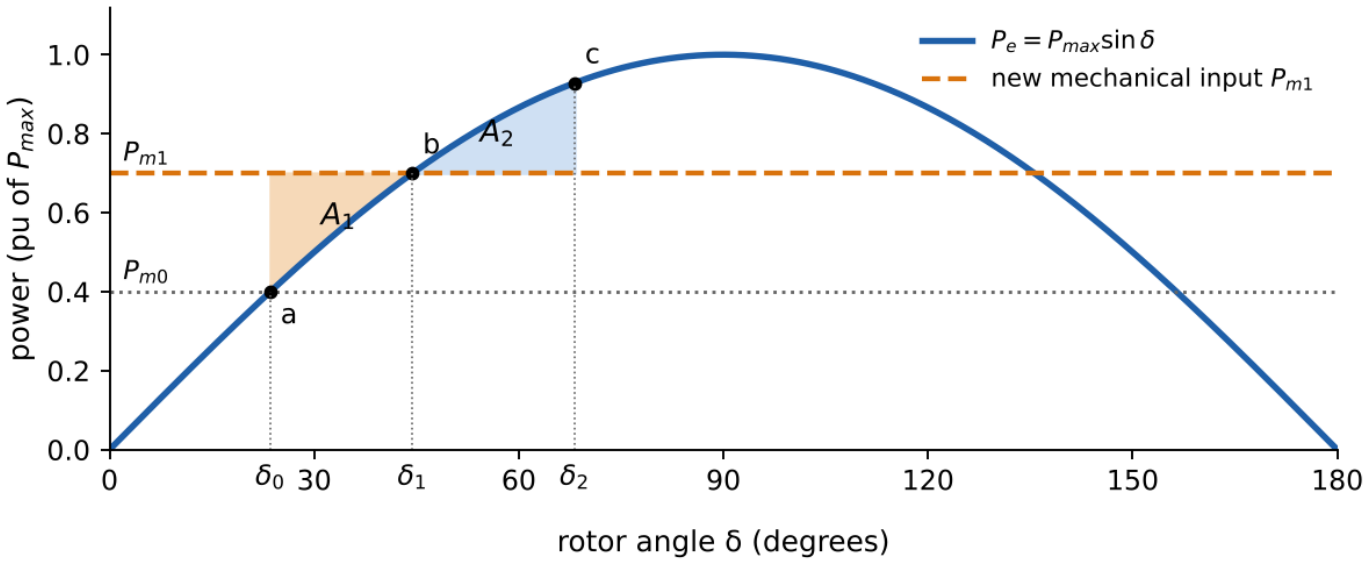


Figure 31: Sudden increase in mechanical input from Pm0 to Pm1: the operating point moves from a, through b, to c; the accelerating area A1 (orange) equals the decelerating area A2 (blue) (reused from the 2021-22 paper)

Let the generator, connected to an infinite bus, run steadily at point a of the power-angle curve $P_e = P_{max}\sin\delta$, with mechanical input $P_{m0} = P_e$ and rotor angle δ_0. The mechanical input is now suddenly raised to P_{m1} (still less than P_{max}).

1. **At the instant of the change.** The rotor angle cannot change instantly, so $\delta = \delta_0$ and $P_e = P_{m0}$. Now $P_{m1} > P_e$: the extra power **accelerates** the rotor above synchronous speed, so δ starts to increase.
2. **Acceleration ($\delta_0 \to \delta_1$).** As δ increases, P_e rises along the curve. The rotor keeps accelerating as long as $P_e < P_{m1}$. The kinetic energy it gains is the **accelerating area** A_1, between the P_{m1} line and the curve, from δ_0 to δ_1.
3. **The new balance point b (δ_1).** Here $P_e = P_{m1}$, so the acceleration is zero. But the rotor is running **above** synchronous speed, so δ keeps increasing: it overshoots.
4. **Deceleration ($\delta_1 \to \delta_2$).** Beyond δ_1, $P_e > P_{m1}$, so the rotor decelerates and gives back energy, the **decelerating area** A_2.
5. **Maximum swing, point c (δ_2).** The rotor is back at synchronous speed when $A_2 = A_1$ (**equal area criterion**); δ is then at its largest.

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6. **Oscillation and settling.** At δ_2, $P_e > P_{m1}$, so the rotor keeps slowing below synchronous speed and δ swings back past δ_1 towards δ_0. Without damping it would swing for ever between δ_0 and δ_2; damping (damper windings, losses) makes the swings die out, and the machine **settles at the new operating point b**, at angle δ_1, delivering P_{m1}.

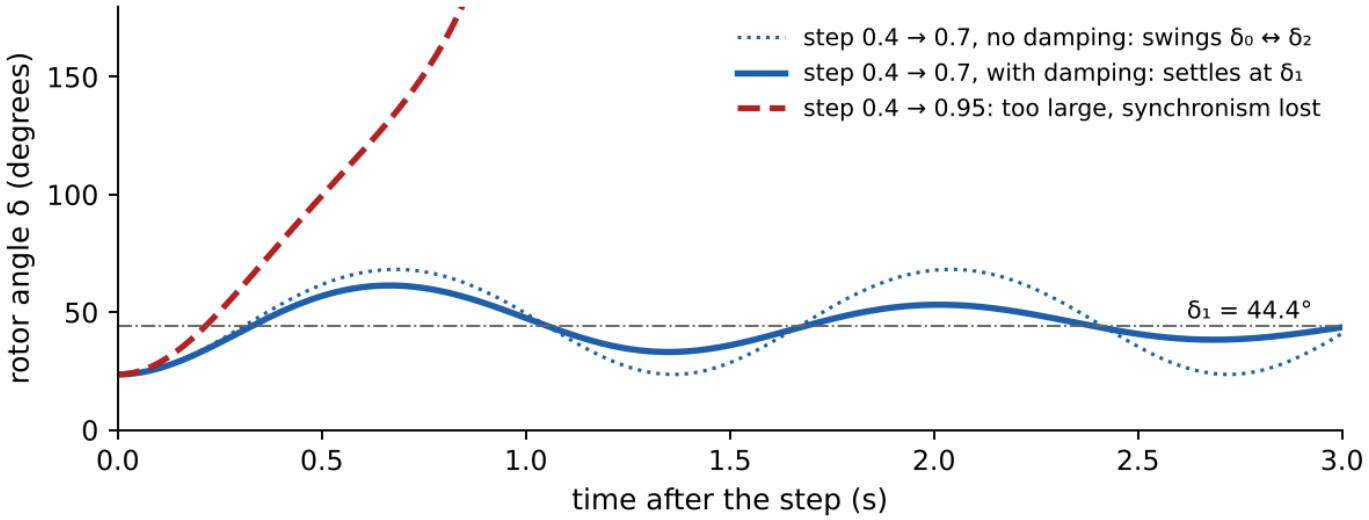


Figure 32: Rotor angle after the step: without damping (dotted blue) it swings between delta0 and delta2; with damping (solid blue) it settles at delta1 = 44.4 degrees; a too large step (dashed red) loses synchronism

Reading the figure. For the step from 0.4 to 0.7 pu of P_{max} (the same as the power-angle diagram above), the dotted blue curve swings between about 24° and 68°; the solid blue curve, with damping, settles at $\delta_1 = 44.4°$. The dashed red curve is a step to 0.95 pu: the accelerating area is larger than the decelerating area available, and δ runs away.

Stability limit. The decelerating area can only extend to $\delta_m = 180° - \delta_1$, where the curve falls below P_{m1} again. If the step is so large that A_1 is bigger than the area available up to δ_m, the rotor passes δ_m, accelerates again and **loses synchronism**. The largest sudden increase the machine can take is the one for which A_2 up to $180° - \delta_1$ just equals A_1: this is the **transient stability limit**, and it is lower than P_{max} (the steady-state limit, which a slow increase could reach).

3. Keyword Explanation

1. Accelerating area (A_1)

The area between P_m and the P_e curve where $P_m > P_e$; the energy stored in the rotor.

2. Transient stability limit

The largest sudden disturbance (here, the largest sudden step of input) after which the machine stays in synchronism.

4. Notes and Remarks

- **Simulated curves:** the time plot was computed from the swing equation with $H = 5$ s and $f = 50$ Hz; the shape, not the exact times, is what matters.
- **Why not 90°?** The machine can lose synchronism after a sudden step even when $P_{m1} < P_{max}$, because of the overshoot.
- The same question is solved in the 2021-22 paper (6(c)), as extra reading.

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6.(c) The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus (|V|=1.0 pu), with the generator terminal voltage (|Vt|=1.0 pu). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (a) System healthy (b) One line shorted (3 phase) in the middle. Marks: 10

1. Theory

1. Per unit, infinite bus and transient reactance (Core Idea)

- **Per unit (pu):** every quantity is written as a fraction of a chosen base value; $|V| = 1.0$ pu simply means rated voltage.
- **Infinite bus:** the large system on the right of Fig. 2, at a fixed $1\angle 0°$ pu (6(a)).
- **Transient reactance X'_d:** for the first second or so after a disturbance, a generator behaves like a constant voltage E' (the **emf behind transient reactance**) in series with X'_d. So for stability studies the generator is drawn as E' behind $X'_d = 0.25$ pu.

2. Power transfer and the transfer reactance

From 6(a), two voltages E' and V joined by a reactance X exchange

$$P = \frac{|E'||V|}{X}\sin\delta, \qquad P_{max} = \frac{|E'||V|}{X}$$

P_{max} depends on the **transfer reactance** X between them. Anything that raises X (opening a line, a fault) lowers P_{max}.

3. Finding E' from the terminal conditions

We know the power and the terminal voltage $|V_t| = 1$, but not its angle. Between V_t and the bus the reactance is that of the transformer and the two lines, so $P = \frac{|V_t||V|}{X}\sin\theta$ gives the angle θ of V_t. Then the current is $I = (V_t - V)/(jX)$, and adding the drop in X'_d gives $E' = V_t + jX'_dI$.

4. Reducing a network with a fault

A three-phase fault connects a point of the network to earth (the zero-voltage reference). The network then has three nodes (E', the infinite bus and earth) and is no longer one series path. A **star–delta conversion** turns it into one reactance between E' and the bus. For a star with arms X_a, X_b, X_c (to the nodes a, b, c), the delta arm between a and b is

$$X_{ab} = \frac{X_aX_b + X_bX_c + X_cX_a}{X_c}$$

2. Answer

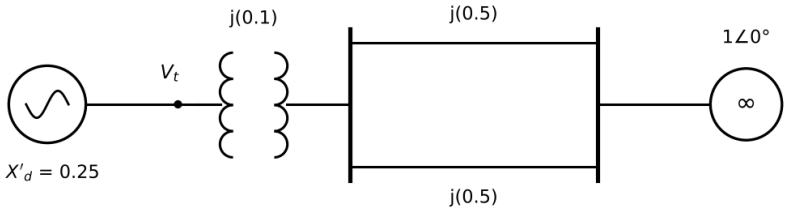


Figure 33: Repeated from the question: Fig. 2 of the paper (redrawn)

Given (Fig. 2): $X'_d = 0.25$, transformer $X_t = 0.1$, two parallel lines of 0.5 each (all pu, reactances); $P = 1.0$ pu; $|V_t| = 1.0$ pu; infinite bus $V = 1\angle 0°$ pu.

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Step 1: Angle of the terminal voltage

Between V_t and the bus: $X = X_t + 0.5 \parallel 0.5 = 0.1 + 0.25 = 0.35$ pu.

$$P = \frac{|V_t||V|}{0.35}\sin\theta \quad \Rightarrow \quad \sin\theta = \frac{1.0 \times 0.35}{1.0 \times 1.0} = 0.35, \qquad \theta = 20.49^\circ$$

$$V_t = 1\angle 20.49^\circ = 0.9367 + j0.3500 \text{ pu}$$

Step 2: Current

$$I = \frac{V_t - V}{jX} = \frac{(0.9367 - 1) + j0.35}{j0.35} = \frac{-0.0633 + j0.35}{j0.35} = 1.0000 + j0.1807 = 1.0162\angle 10.24^\circ \text{ pu}$$

Step 3: EMF behind transient reactance

$$E' = V_t + jX'_dI = (0.9367 + j0.35) + j0.25(1.0 + j0.1807)$$

$$E' = (0.9367 - 0.0452) + j(0.35 + 0.25) = 0.8916 + j0.6000 = 1.0747\angle 33.94^\circ \text{ pu}$$

$$\boxed{E' = 1.0747\angle 33.94^\circ \text{ pu}}$$

So the initial rotor angle is $\delta_0 = 33.94^\circ$.

Step 4: (a) System healthy

$$X_I = X'_d + X_t + \frac{0.5}{2} = 0.25 + 0.1 + 0.25 = 0.60 \text{ pu}$$

$$P_{max,I} = \frac{|E'||V|}{X_I} = \frac{1.0747 \times 1.0}{0.60} = 1.791 \text{ pu}$$

$$\boxed{P_{max} = 1.791 \text{ pu (healthy)}}$$

Check: $1.791 \sin 33.94^\circ = 1.0$ pu, the power being delivered.

Step 5: (b) Three-phase short at the middle of one line

The faulted line splits into two halves of $j0.25$, with the fault point F at earth potential.

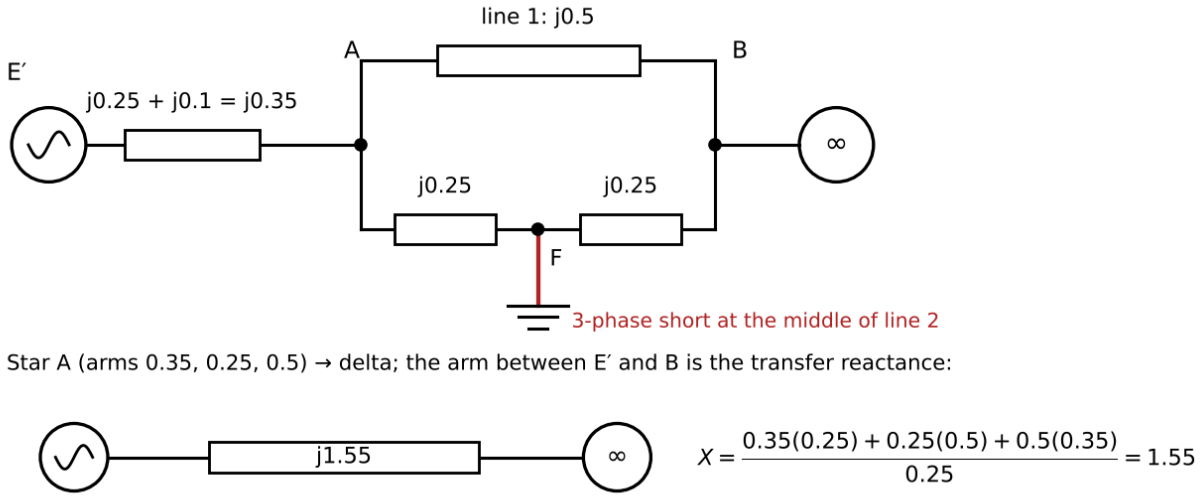


Figure 34: Network with the 3-phase short at the middle of line 2, and its reduction to one transfer reactance of j1.55 (reused from the 2021-22 paper)

At node A (the sending end of the lines) there is a star of three reactances:

- to E': $X_a = 0.25 + 0.1 = 0.35$

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- to the fault (earth): $X_c = 0.25$ (half of line 2)
- to the infinite bus B: $X_b = 0.5$ (line 1)

The other half of line 2 (from B to F, $j0.25$) hangs from the infinite bus to earth; it carries current from the bus but does not affect the power from E'. Converting the star at A into a delta, the arm between E' and B is

$$X_{II} = \frac{X_a X_b + X_b X_c + X_c X_a}{X_c} = \frac{0.35 \times 0.5 + 0.5 \times 0.25 + 0.25 \times 0.35}{0.25} = \frac{0.3875}{0.25} = 1.55 \text{ pu}$$

$$P_{max,II} = \frac{1.0747 \times 1.0}{1.55} = 0.693 \text{ pu}$$

$$\boxed{P_{max} = 0.693 \text{ pu (mid-line fault)}}$$

This is **less than the 1.0 pu input**: during the fault the generator cannot deliver its mechanical input, so it accelerates, and it stays in step only if the fault is cleared quickly enough (6(a)).

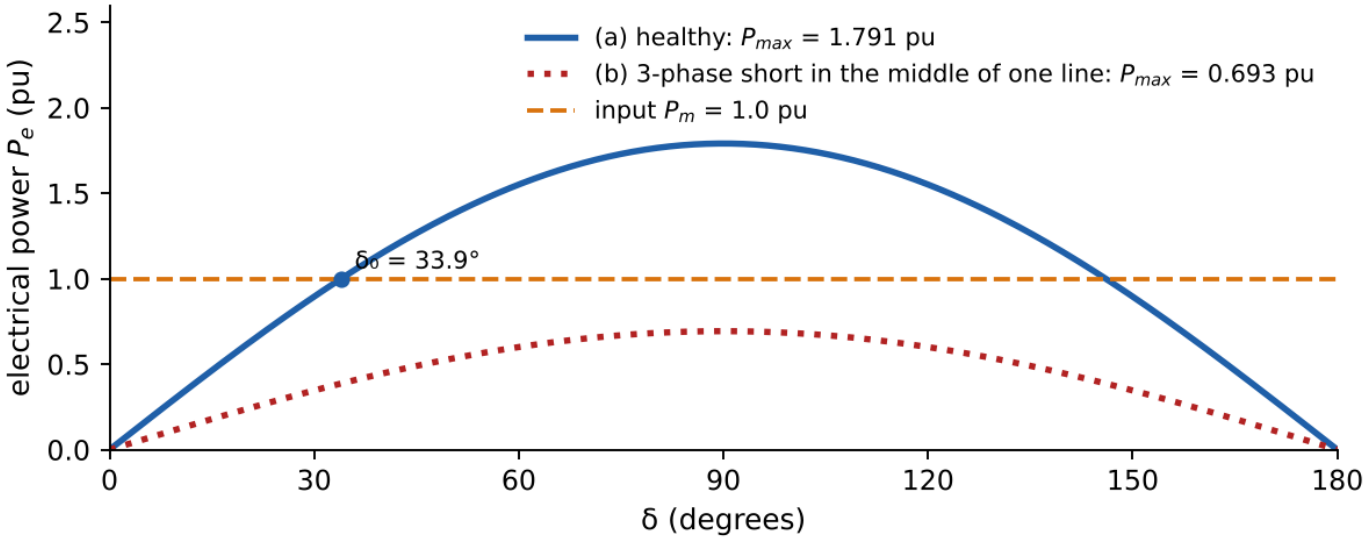


Figure 35: Power-angle curves: healthy (solid blue) and with the mid-line fault (dotted red), with the 1.0 pu input (dashed orange)

Reading the figure. The healthy curve meets the 1.0 pu input at $\delta_0 = 33.9°$; the fault curve, with a peak of only 0.693 pu, never reaches it.
Final Answer: $E' = 1.0747\angle 33.94°$ pu; $P_{max} = 1.791$ pu with the system healthy and 0.693 pu with a three-phase short at the middle of one line.

3. Keyword Explanation

1. Emf behind transient reactance (E')

The constant internal voltage that represents the generator for the first swings after a disturbance.

2. Transfer reactance

The single reactance between the generator emf and the infinite bus after the network has been reduced; $P_{max} = E'V/X$.

4. Notes and Remarks

- **Why the fault hurts so much:** the short at F takes much of the generator's current to earth, so the coupling to the infinite bus is weak: the transfer reactance rises from 0.60 to 1.55 pu.
- **After the fault is cleared** by opening the faulted line, $X = 0.25 + 0.1 + 0.5 = 0.85$ pu and $P_{max} = 1.264$ pu; this curve is used for the equal area example in 6(a).

- The same network, with a third case (one line open) and the plot of all three curves, is solved in 6(d) of the 2021-22 paper, as extra reading.

5. Final Concise Answer

(a) The power angle δ is the angle between the generator's internal emf and the infinite-bus voltage; $P_e = (EV/X)\sin\delta$. Equal area criterion: after a disturbance the machine stays stable if the decelerating area ($P_e > P_m$) can equal the accelerating area ($P_m > P_e$), $\int_{\delta_0}^{\delta_{max}} (P_m - P_e)\, d\delta = 0$; with the 6(c) numbers the critical clearing angle is 59.7°.

(b) A sudden rise from P_{m0} to P_{m1} accelerates the rotor ($P_m > P_e$); δ passes the new equilibrium δ_1, overshoots to δ_2 where $A_2 = A_1$, then oscillates and, with damping, settles at δ_1. If A_1 is larger than the area available up to $180° - \delta_1$, synchronism is lost.

(c) $V_t = 1\angle 20.49°$; $I = 1.0162\angle 10.24°$; $E' = 1.0747\angle 33.94°$ pu; $P_{max} = 1.791$ pu (healthy, $X = 0.60$) and 0.693 pu (3-phase short at the middle of one line, $X = 1.55$).

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FORMULA SHEET

EEE337

Power System II

Formula Sheet

Session 2020–21 paper

Every formula behind the 3rd Year 2nd Semester Final Examination, 2024

TOPICS

1 Inductance of lines — 17 formulas

2 Capacitance of lines and corona — 13 formulas

3 Insulators and supports — 11 formulas

4 Cables — 14 formulas

5 Power flow and stability — 11 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

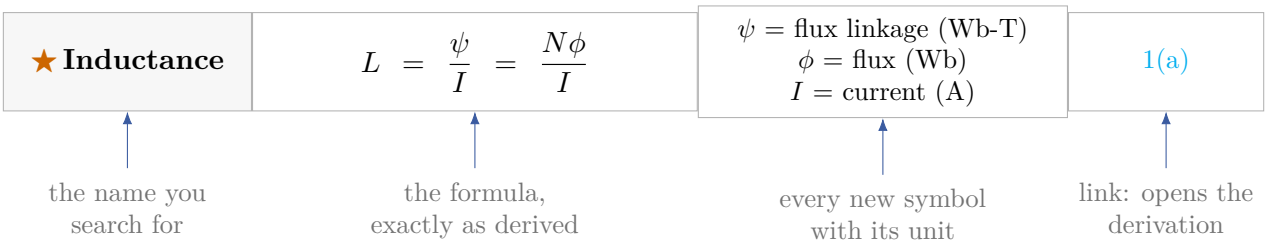
Prepared by Nur Mohammad

October 2026

BACK OF THE COVER

Before you use the sheet

How to read a row



In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Inductance of lines 4

2 Capacitance of lines and corona 5

3 Insulators and supports 6

4 Cables 6

5 Power flow and stability 7

6 Key formulas 9

7 Symbol index 11

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Conventions (V.K. Mehta; Grainger & Stevenson for ACSR data, bundled lines and stability; Hingorani & Gyugyi for FACTS): $\mu_0 = 4\pi \times 10^{-7}$ H/m and $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m; ln is the natural log; line voltages are line-to-line unless marked V_{ph}; r is the conductor radius, D_s its GMR and D a centre-to-centre spacing (same unit); in a cable d is the core diameter and D the inside diameter of the sheath; K is the ratio of shunt to self capacitance of a string unit. References point to the solutions of the 2020–21 paper.

Columns: Title · Equation · Notation · Ref. (reference).

1. Inductance of lines:

Title	Equation	Notation	Ref.
★ **Inductance**	$L = \dfrac{\psi}{I} = \dfrac{N\phi}{I}$	ψ = flux linkage (Wb-T) ϕ = flux (Wb) I = current (A)	1(a)
Induced voltage	$e = L\dfrac{di}{dt}$	e = self-induced voltage (V)	1(a)
Internal flux linkage	$\psi_{int} = \dfrac{\mu_0 I}{8\pi}$	uniform current in a solid round conductor	1(b)
External flux linkage	$\psi_{12} = \dfrac{\mu_0 I}{2\pi} \ln \dfrac{D_2}{D_1} = 2 \times 10^{-7}\, I \ln \dfrac{D_2}{D_1}$	D_1, D_2 = distances of two points from the conductor (m)	1(b)
★ **GMR of a solid conductor**	$D_s = r' = r\, e^{-1/4} = 0.7788\, r$	for ACSR use the table value (Pheasant: 0.0466 ft)	1(b)
Flux linkage in a group	$\psi_a = 2 \times 10^{-7} \left[I_a \ln \dfrac{1}{D_s} + I_b \ln \dfrac{1}{D_{ab}} + I_c \ln \dfrac{1}{D_{ac}} \right]$	$I_a + I_b + I_c = 0$	2(b)
★ **Transposed 3-phase line**	$L = 2 \times 10^{-7} \ln \dfrac{D_{eq}}{D_s}$ H/m	per phase	1(b)
★ **Equivalent spacing**	$D_{eq} = \sqrt[3]{D_{ab} D_{bc} D_{ca}}$	flat spacing $D, D, 2D$: $D_{eq} = 1.26\, D$	1(b)
★ **Two-conductor bundle**	$D_s^b = \sqrt{D_s\, d}$	d = spacing of the sub-conductors (m)	1(b)
★ **Inductive reactance**	$X_L = 2\pi f L$	f = frequency (Hz); 1 mile = 1.609 km	1(b)
★ **Composite conductor**	$L_X = 2 \times 10^{-7} \ln \dfrac{D_m}{D_{sX}}$ H/m	D_m = mutual GMD between X and Y (m) D_{sX} = self GMD of X (m)	2(c)
Mutual GMD	$D_m = \sqrt[mn]{(D_{aa'} D_{ab'} \cdots D_{am}) \cdots (D_{na'} \cdots D_{nm})}$	n, m = strands in X and Y	2(c)
Self GMD	$D_{sX} = \sqrt[n^2]{(D_{aa} D_{ab} \cdots D_{an}) \cdots (D_{na} \cdots D_{nn})}$	$D_{aa} = r'_a$, $D_{bb} = r'_b$, …	2(c)
Strand inductance	$L_X = \dfrac{L_a + L_b + \cdots + L_n}{n^2}$	n strands in parallel	2(c)
Composite line loop	$L = 2 \times 10^{-7} \ln \dfrac{D_m^2}{D_{sX} D_{sY}}$ H/m	both sides of a single-phase line	2(c)

Title	Equation	Notation	Ref.
★ **Mutual inductance**	$M = 2 \times 10^{-7} \ln \dfrac{D_{P_1T_2}\, D_{P_2T_1}}{D_{P_1T_1}\, D_{P_2T_2}}$ H/m	$D_{P_iT_j}$ = distance from power wire i to telephone wire j (m)	1(d)
Induced voltage	$E = 2\pi f M I$	E = voltage in the telephone line (V per m)	1(d)

Table 1: Inductance of lines

2. Capacitance of lines and corona:

Title	Equation	Notation	Ref.
Capacitance	$C = \dfrac{q}{V}$	q = charge (C/m)	1(c)
Field of a line charge	$E = \dfrac{q}{2\pi\varepsilon_0\, x}$	x = distance from the conductor (m)	1(c)
Potential difference	$V_{12} = \dfrac{q}{2\pi\varepsilon_0} \ln \dfrac{D_2}{D_1}$	between two points at D_1 and D_2	1(c)
Balanced voltages	$V_{ab} + V_{ac} = 3V_{an}$	V_{an} = phase voltage	1(c)
★ **Capacitance to neutral**	$C_n = \dfrac{2\pi\varepsilon_0}{\ln(D_{eq}/r)} = \dfrac{0.0556}{\ln(D_{eq}/r)}$ μF/km	transposed 3-phase line, per phase; outside radius r	1(c)
Line charging current	$I_c = 2\pi f\, C_n\, V_{ph}$	V_{ph} = phase voltage (V)	1(c)
Field near the conductor	$g = \dfrac{V}{x\, \ln(d/r)}$	V = phase-to-neutral voltage (kV) d = spacing (cm)	2(a)
Strength of air	$g_0 = 30$ kV/cm peak $= 21.2$ kV/cm r.m.s.	at 76 cm Hg and 25 °C	2(a)
Surface field	$g_{max} = \dfrac{V}{r\, \ln(d/r)}$	largest at the conductor surface	4(a)
★ **Critical disruptive voltage**	$V_c = m_0\, g_0\, \delta\, r\, \ln \dfrac{d}{r}$ kV	m_0 = irregularity factor (1 smooth, 0.80–0.87 stranded) r = conductor radius (cm)	4(a)
Air density factor	$\delta = \dfrac{3.92\, b}{273 + t}$	b = pressure (cm of Hg) t = temperature (°C)	4(a)
Visual critical voltage	$V_v = m_v\, g_0\, \delta\, r \left(1 + \dfrac{0.3}{\sqrt{\delta r}}\right) \ln \dfrac{d}{r}$	m_v = irregularity factor for visual corona	4(a)
Skin depth	$\delta = \dfrac{1}{\sqrt{\pi f \mu \sigma}}$	σ = conductivity (S/m); about 9 mm for copper at 50 Hz	3(a)

Table 2: Capacitance of lines and corona

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3. Insulators and supports:

Title	Equation	Notation	Ref.
★ **Capacitance ratio**	$K = \dfrac{C_1}{C}$	C_1 = shunt capacitance to the tower (F) C = self capacitance of a unit (F)	4(b)
★ **Unit voltages**	$V_2 = (1+K)V_1,\ V_3 = (1+3K+K^2)V_1$	V_1 = voltage of the top (tower) unit	4(b)
General unit	$V_{k+1} = V_k + K\,(V_1 + V_2 + \cdots + V_k)$	node equation at the link below unit k	4(b)
★ **String efficiency**	$\eta = \dfrac{V}{n\,V_n} \times 100\ \%$	V = phase voltage across the string n = number of units V_n = voltage of the unit nearest the line	4(b)
Three-unit string	$\eta = \dfrac{3+4K+K^2}{3(1+3K+K^2)}$	$K = 0.1$: 86.8 %	4(b)
★ **Capacitance grading**	$C_2 = C_1(1+K),\ C_3 = C_1(1+3K)$	disc capacitances for equal voltages, top unit C_1	4(b)
Safety factor of an insulator	$\text{SF} = \dfrac{\text{puncture strength}}{\text{flash-over voltage}}$	about 10	5(a)
★ **Working tension**	$T = \dfrac{\text{ultimate strength}}{\text{safety factor}}$	kg-wt or N	5(c)
★ **Sag, level supports**	$S = \dfrac{w\,l^2}{8\,T}$	w = weight per metre l = span (m) T = tension (same force unit as w)	5(c)
Conductor curve	$y = \dfrac{w\,x^2}{2\,T}$	x = distance from the lowest point (m)	5(c)
★ **Height of supports**	$h = \text{ground clearance} + S$	supports at the same level	5(c)

Table 3: Insulators and supports

4. Cables:

Title	Equation	Notation	Ref.
Flux density in the dielectric	$D_x = \dfrac{Q}{2\pi x}$	Q = charge on the core (C/m) x = distance from the centre (m)	3(c)
Stress from the charge	$g_x = \dfrac{Q}{2\pi\varepsilon_0\varepsilon_r\,x}$	ε_r = relative permittivity	3(c)
Core-to-sheath voltage	$V = \dfrac{Q}{2\pi\varepsilon_0\varepsilon_r}\ln\dfrac{D}{d}$	d = core diameter, D = inside diameter of the sheath	3(c)
★ **Dielectric stress**	$g_x = \dfrac{V}{x\,\ln(D/d)}$	falls as $1/x$	3(c)

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Title	Equation	Notation	Ref.
★ **Maximum and minimum stress**	$g_{max} = \frac{2V}{d\,\ln(D/d)},\ g_{min} = \frac{2V}{D\,\ln(D/d)}$	at the core and at the sheath	2(d)
★ **Stress ratio**	$\frac{g_{max}}{g_{min}} = \frac{D}{d}$	gives D from the two stresses	4(c)
Insulation thickness	$t = \frac{D-d}{2}$		4(c)
Operating voltage	$V = g_{max}\,\frac{d}{2}\ln\frac{D}{d}$	core-to-sheath, r.m.s. if g is r.m.s.	4(c)
Cable capacitance	$C = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(D/d)}$ F/m	single-core cable	3(c)
★ **Belted cable tests**	$C_1 = 3C_s,\ C_2 = C_s + 2C_c$	C_s = core to sheath C_c = core to core test 1: cores bunched to sheath test 2: one core to the rest + sheath	3(d)
★ **Capacitance per phase**	$C_N = C_s + 3C_c$	the delta of three C_c acts as a star of $3C_c$	3(d)
★ **Cable charging current**	$I_c = 2\pi f\,C_N\,V_{ph}$	$V_{ph} = V_L/\sqrt{3}$	3(d)
★ **Varley loop balance**	$\frac{P}{Q} = \frac{R_{loop}-R_X}{R_X+S},\ R_X = \frac{Q\,R_{loop}-P\,S}{P+Q}$	S = variable resistance in series with the faulty conductor (Ω) R_{loop} = faulty + healthy conductor (Ω)	5(b)
Equal ratio arms	$R_X = \frac{R_{loop}-S}{2}$	$P = Q$; distance $= R_X/(\Omega$ per km$)$	5(b)

Table 4: Cables

5. Power flow and stability:

Title	Equation	Notation	Ref.
★ **Power over a line**	$P = \frac{V_1V_2}{X}\sin\delta$	what FACTS controllers change: V (shunt), X (series), all three (UPFC)	3(b)
★ **Power-angle equation**	$P_e = \frac{\lvert E\rvert\lvert V\rvert}{X}\sin\delta = P_{max}\sin\delta$	δ = power angle (E ahead of V) P_{max} at $\delta = 90°$	6(a)
Swing equation	$M\frac{d^2\delta}{dt^2} = P_m - P_e$	M = inertia constant P_m = mechanical input (pu)	6(a)
★ **Equal area criterion**	$\int_{\delta_0}^{\delta_{max}}(P_m - P_e)\,d\delta = 0 \Leftrightarrow A_1 = A_2$	A_1 = accelerating area A_2 = decelerating area	6(a)
Critical clearing angle	$\cos\delta_{cr} = \frac{P_m(\delta_{max}-\delta_0) + P_{3max}\cos\delta_{max} - P_{2max}\cos\delta_0}{P_{3max}-P_{2max}}$	P_{2max} during, P_{3max} after the fault; angles in rad	6(a)

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Title	Equation	Notation	Ref.
Limit of the swing	$\delta_{max} = 180° - \delta_1$	δ_1 = new operating angle	6(b)
Terminal voltage angle	$\sin\theta = \dfrac{P\,X}{\lvert V_t\rvert\lvert V\rvert}$	X = reactance from V_t to the bus (pu)	6(c)
Current	$I = \dfrac{V_t - V}{jX}$	pu	6(c)
★ **Transient emf**	$E' = V_t + jX'_dI$	X'_d = transient reactance (pu)	6(c)
★ **Maximum power**	$P_{max} = \dfrac{\lvert E'\rvert\lvert V\rvert}{X}$	X = transfer reactance (pu)	6(c)
Transfer reactance (Y to Δ)	$X_{ab} = \dfrac{X_aX_b + X_bX_c + X_cX_a}{X_c}$	X_c = branch to the faulted node (pu)	6(c)

Table 5: Power flow and stability

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6. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Inductance	$L = \frac{\psi}{I} = \frac{N\phi}{I}$
GMR of a solid conductor	$D_s = r' = r\,e^{-1/4} = 0.7788\,r$
Transposed 3-phase line	$L = 2\times10^{-7}\ln\frac{D_{eq}}{D_s}$ H/m
Equivalent spacing	$D_{eq} = \sqrt[3]{D_{ab}D_{bc}D_{ca}}$
Two-conductor bundle	$D_s^b = \sqrt{D_s\,d}$
Inductive reactance	$X_L = 2\pi f L$
Composite conductor	$L_X = 2\times10^{-7}\ln\frac{D_m}{D_{sX}}$ H/m
Mutual inductance	$M = 2\times10^{-7}\ln\frac{D_{P_1T_2}\,D_{P_2T_1}}{D_{P_1T_1}\,D_{P_2T_2}}$ H/m
Capacitance to neutral	$C_n = \frac{2\pi\varepsilon_0}{\ln(D_{eq}/r)} = \frac{0.0556}{\ln(D_{eq}/r)}$ μF/km
Critical disruptive voltage	$V_c = m_0\,g_0\,\delta\,r\,\ln\frac{d}{r}$ kV
Capacitance ratio	$K = \frac{C_1}{C}$
Unit voltages	$V_2 = (1+K)V_1,\ V_3 = (1+3K+K^2)V_1$
String efficiency	$\eta = \frac{V}{n\,V_n}\times100\ \%$
Capacitance grading	$C_2 = C_1(1+K),\ C_3 = C_1(1+3K)$
Working tension	$T = \frac{\text{ultimate strength}}{\text{safety factor}}$
Sag, level supports	$S = \frac{w\,l^2}{8\,T}$
Height of supports	$h = \text{ground clearance} + S$
Dielectric stress	$g_x = \frac{V}{x\,\ln(D/d)}$
Maximum and minimum stress	$g_{max} = \frac{2V}{d\,\ln(D/d)},\ g_{min} = \frac{2V}{D\,\ln(D/d)}$
Stress ratio	$\frac{g_{max}}{g_{min}} = \frac{D}{d}$
Belted cable tests	$C_1 = 3C_s,\ C_2 = C_s + 2C_c$
Capacitance per phase	$C_N = C_s + 3C_c$

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Title	Equation
Cable charging current	$I_c = 2\pi f\, C_N\, V_{ph}$
Varley loop balance	$\frac{P}{Q} = \frac{R_{loop} - R_X}{R_X + S},\ R_X = \frac{Q\, R_{loop} - P\, S}{P + Q}$
Power over a line	$P = \frac{V_1 V_2}{X} \sin\delta$
Power-angle equation	$P_e = \frac{\lvert E\rvert\lvert V\rvert}{X} \sin\delta = P_{max} \sin\delta$
Equal area criterion	$\int_{\delta_0}^{\delta_{max}} (P_m - P_e)\, d\delta = 0 \ \Leftrightarrow\ A_1 = A_2$
Transient emf	$E' = V_t + jX'_d I$
Maximum power	$P_{max} = \frac{\lvert E'\rvert\lvert V\rvert}{X}$

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7. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
ψ	flux linkage (Wb-T)	Inductance
ϕ	flux (Wb)	Inductance
I	current (A)	Inductance
e	self-induced voltage (V)	Induced voltage
D_1, D_2	distances of two points from the conductor (m)	External flux linkage
d	spacing of the sub-conductors (m)	Two-conductor bundle
f	frequency (Hz); 1 mile = 1.609 km	Inductive reactance
D_m	mutual GMD between X and Y (m)	Composite conductor
D_{sX}	self GMD of X (m)	Composite conductor
n, m	strands in X and Y	Mutual GMD
$D_{P_iT_j}$	distance from power wire i to telephone wire j (m)	Mutual inductance
E	voltage in the telephone line (V per m)	Induced voltage
q	charge (C/m)	Capacitance
x	distance from the conductor (m)	Field of a line charge
V_{an}	phase voltage	Balanced voltages
V_{ph}	phase voltage (V)	Line charging current
V	phase-to-neutral voltage (kV)	Field near the conductor
m_0	irregularity factor (1 smooth, 0.80–0.87 stranded)	Critical disruptive voltage
r	conductor radius (cm)	Critical disruptive voltage
b	pressure (cm of Hg)	Air density factor
t	temperature (°C)	Air density factor
m_v	irregularity factor for visual corona	Visual critical voltage
σ	conductivity (S/m); about 9 mm for copper at 50 Hz	Skin depth
C_1	shunt capacitance to the tower (F)	Capacitance ratio
C	self capacitance of a unit (F)	Capacitance ratio
V_1	voltage of the top (tower) unit	Unit voltages
n	number of units	String efficiency
V_n	voltage of the unit nearest the line	String efficiency
w	weight per metre	Sag, level supports
l	span (m)	Sag, level supports
T	tension (same force unit as w)	Sag, level supports
Q	charge on the core (C/m)	Flux density in the dielectric
ε_r	relative permittivity	Stress from the charge
C_s	core to sheath	Belted cable tests
C_c	core to core	Belted cable tests
S	variable resistance in series with the faulty conductor (Ω)	Varley loop balance

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Symbol	Meaning	First used in
R_{loop}	faulty + healthy conductor (Ω)	Varley loop balance
δ	power angle (E ahead of V)	Power-angle equation
M	inertia constant	Swing equation
P_m	mechanical input (pu)	Swing equation
A_1	accelerating area	Equal area criterion
A_2	decelerating area	Equal area criterion
δ_1	new operating angle	Limit of the swing
X	reactance from V_t to the bus (pu)	Terminal voltage angle
X'_d	transient reactance (pu)	Transient emf
X_c	branch to the faulted node (pu)	Transfer reactance (Y to Δ)

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